A SURVEY OF PASSIVE MAGNETIC SPRINGS

40th ASPE Annual Meeting
San Diego, California
2025 November 05

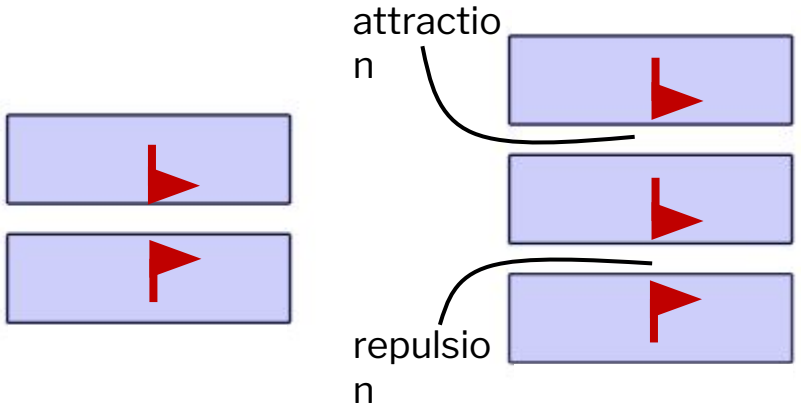
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Byron Seaberg[2]
Yoonsik Kwon[2]
Kareem S. Nayfeh[3]
David L. Trumper[2]
Samir A. Nayfeh[1]

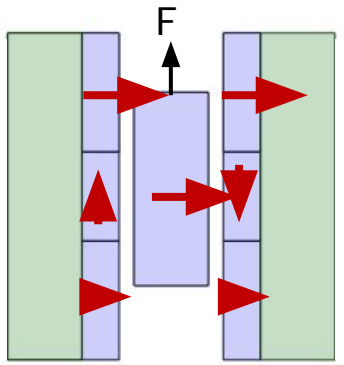
[1]Physical Precision LLC
[2]MIT Precision Motion Control Lab
[3]George Mason University

Creating constant forces with magnets

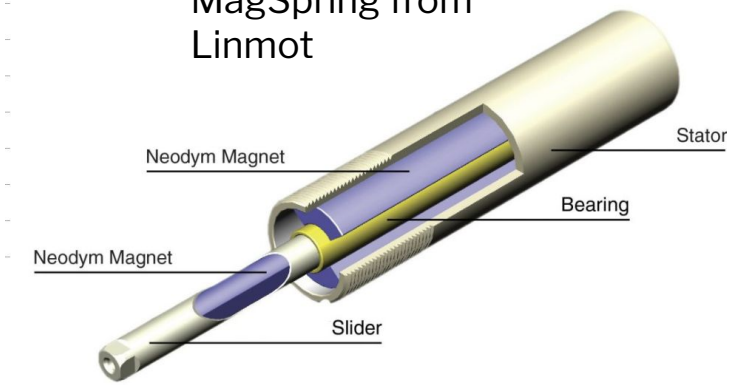
- Motivation: Tractable formulas to design

- What are the goals for designing a magnet spring?
  - Force, stiffness, range of travel
  - Repeatability, low hysteresis, damping, parasitic forces/stiffness, temperature effects, packaging
  - How can this be designed with relatively simple equations, without jumping straight to 3D FEA?
    - How big?
    - Iron or iron-less
    - Halbach magnets?
    - Hysteresis effects?
    - Optimal aspect ratios and gaps?





MagSpring from Linmot



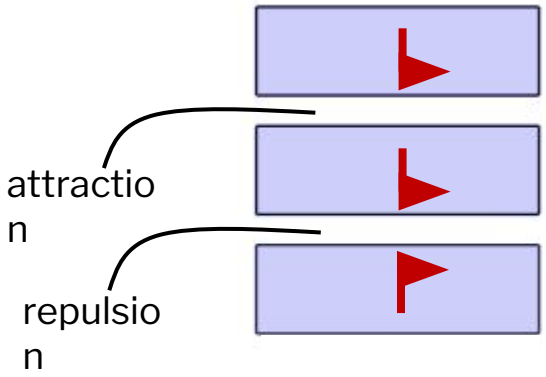
Applications of magnet springs and constant force generators

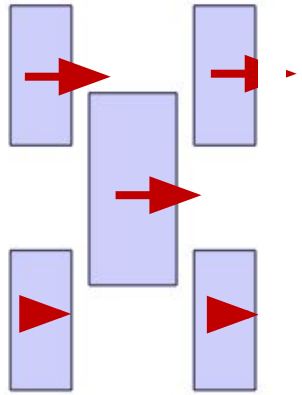
- Gravity compensation
  - Reduced energy consumption
  - Reduced actuator size
- Vibration isolation
  - Quasi-zero stiffness springs
- Non-contact
  - Wear, friction, cleanliness
  - Minimal damping (hysteresis and eddy currents)

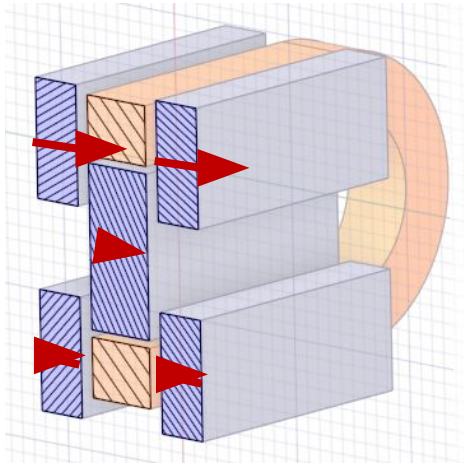
- Alternatives are mechanical springs or pneumatics

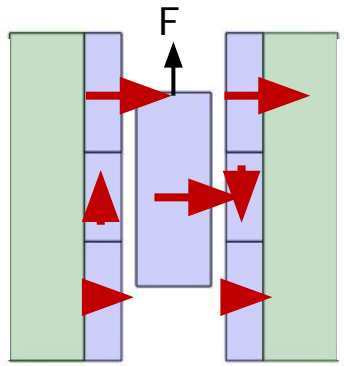
Survey of magnet spring papers – configurations

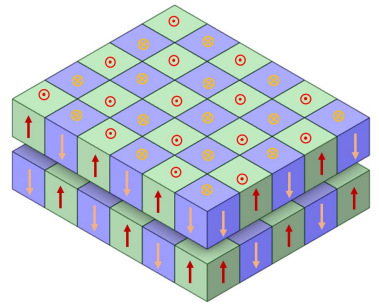
- Configurations
  - Normal – horizontal gap
    - JLG Janssen 2013, Zhang et al 2023
  - Shear – vertical gap
    - JLG Janssen 2011, Berkhof 2015
  - Torsional
    - Boisclair and Richard 2017, Kozakiewicz and Winiarski 2023, Mrak et al 2019
  - Actuators with gravity compensation Wang 2015, Deng 2017, Westhoff and Maas 2024, Jansen 2007
- Modifications
  - Arrays, Iron, Ironless, Halbach
- Optimization
  - Mrak 2022, Zhang et al 2019, Robertson 2011, Hol 2004
- Survey papers
  - Zhu and Chai 2024, Zhou and Wu 2022





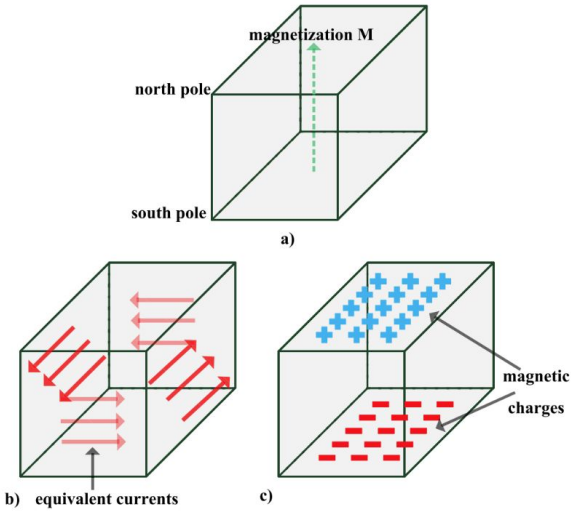




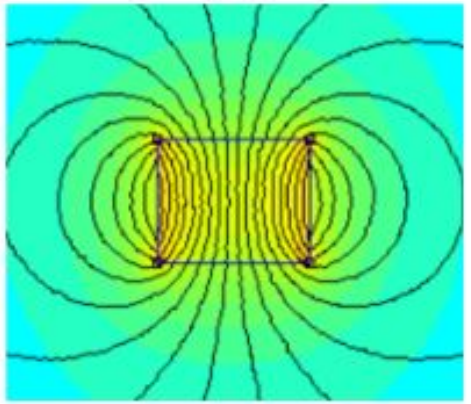


Survey of magnet spring papers – analysis methods

- Analysis methods
  - Analytical
    - Frequently cumbersome equations
      - Pairs of magnets
    - Equivalent charge or current
      - Harmonic model (Fourier series)
    - Lumped-parameter circuit
      - Reluctance is analogous to an electrical resistor
      - MMF ~ voltage
      - Flux ~ current
  - 2D FEA – FEMM is free, open source
  - 3D FEA
  - Semi-analytical

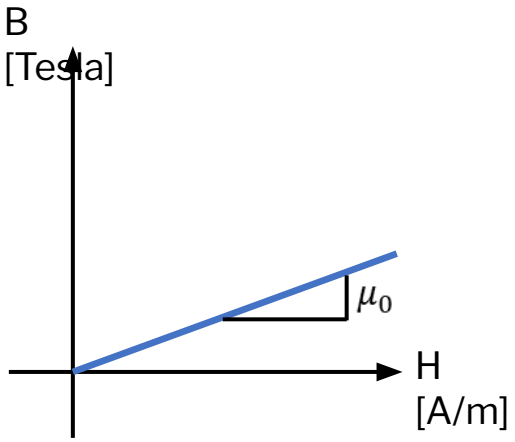


From Lahdo et al 2017

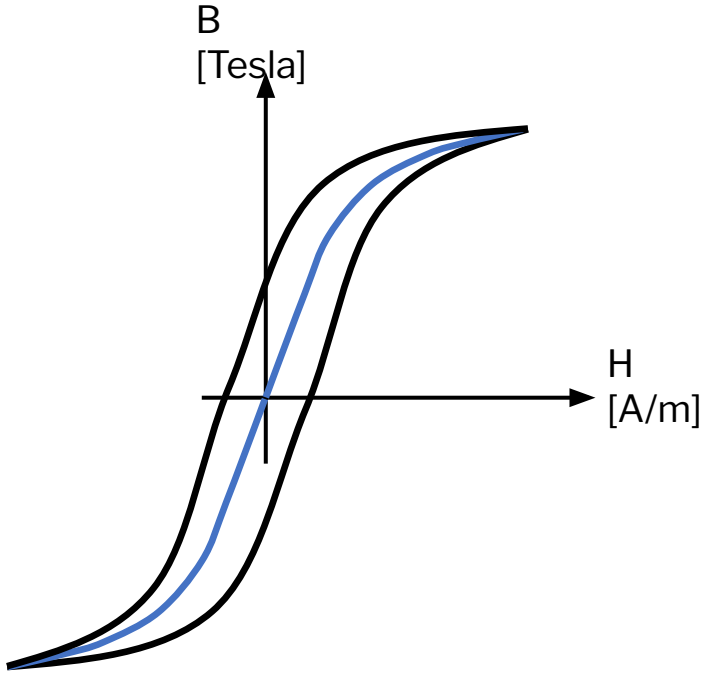


Quick refresher on magnetic materials

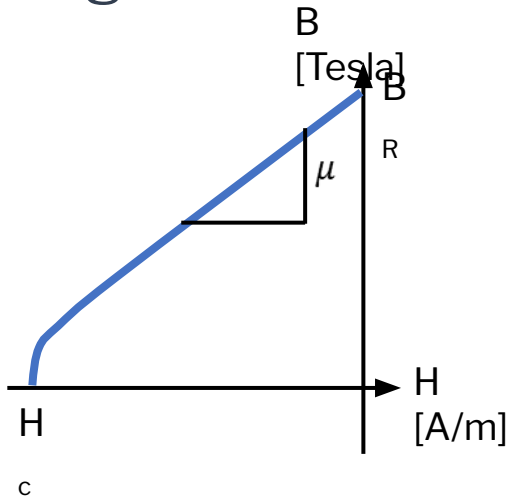
- Linear material – vacuum or air



- Soft magnetic material – Iron



- Magnet – hard magnetic material



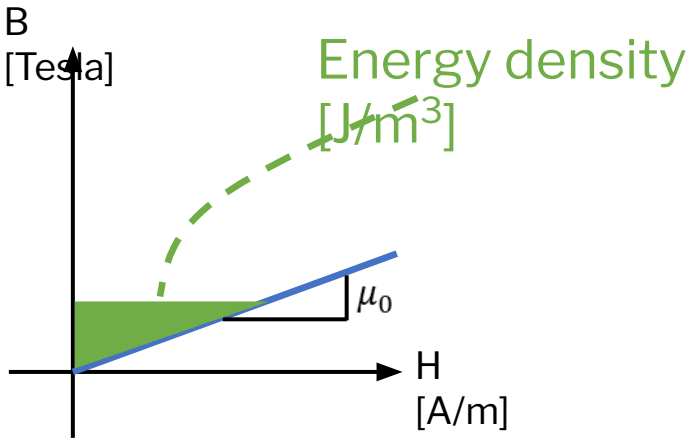
$$B = B_r + \mu H$$

B: magnetic flux density [Tesla]
H: magnetic field intensity [A/m]

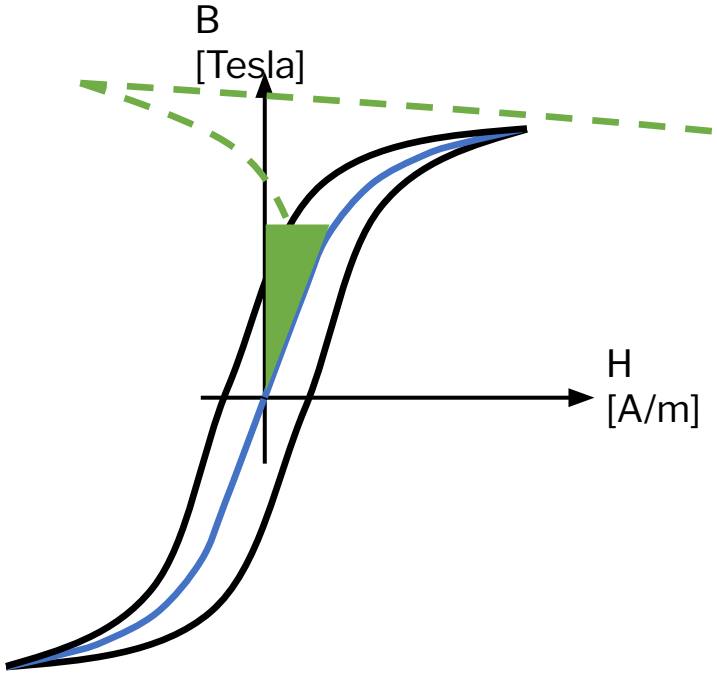
Permeability:
For a linear, non-dispersive material $B = \mu_0 H$
Free space $\mu_0 = 4\pi \times 10^{-7}$ T-m/A
Relative permeability $\mu_R = \mu/\mu_0$

Quick refresher on magnetic materials

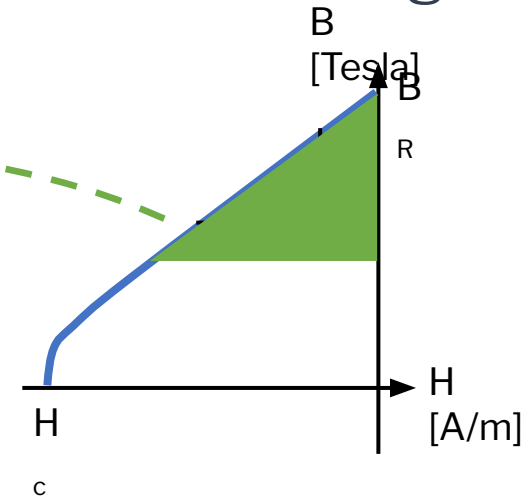
- Linear material – vacuum or air



- Soft magnetic material – Iron



- Hard magnetic material – magnet

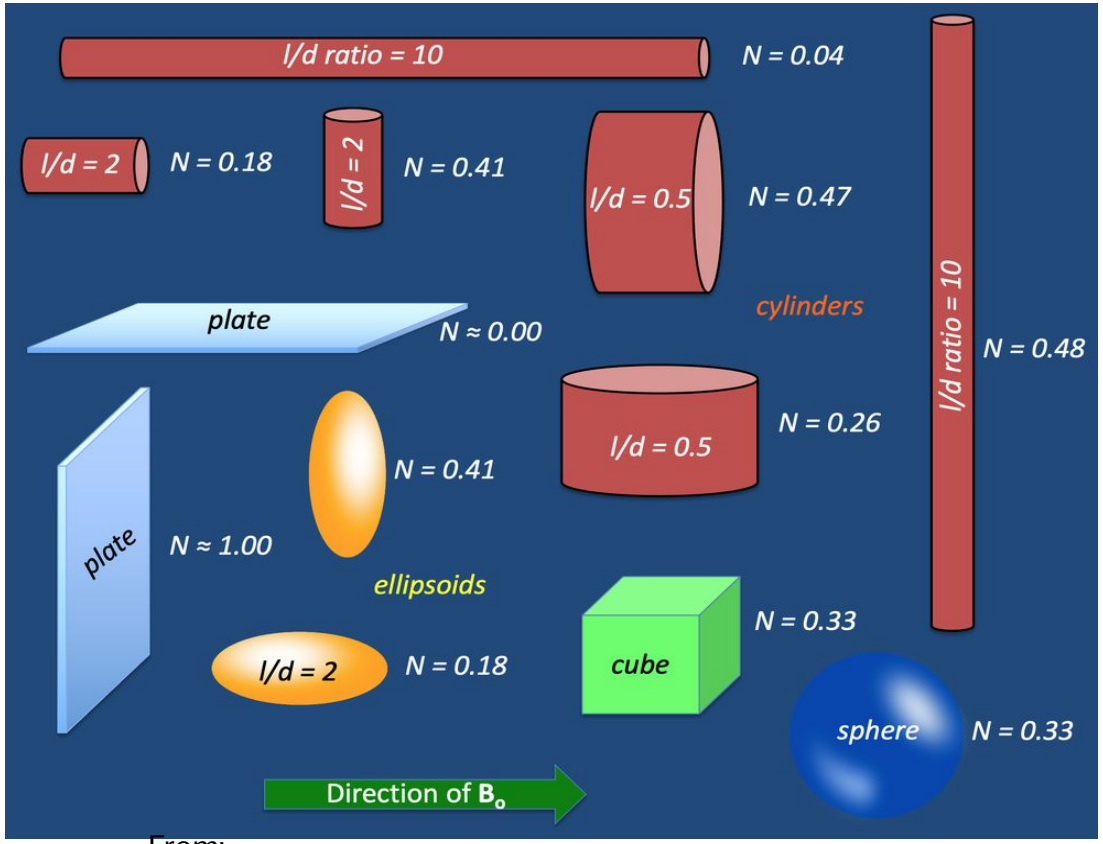


B: magnetic flux density [Tesla]
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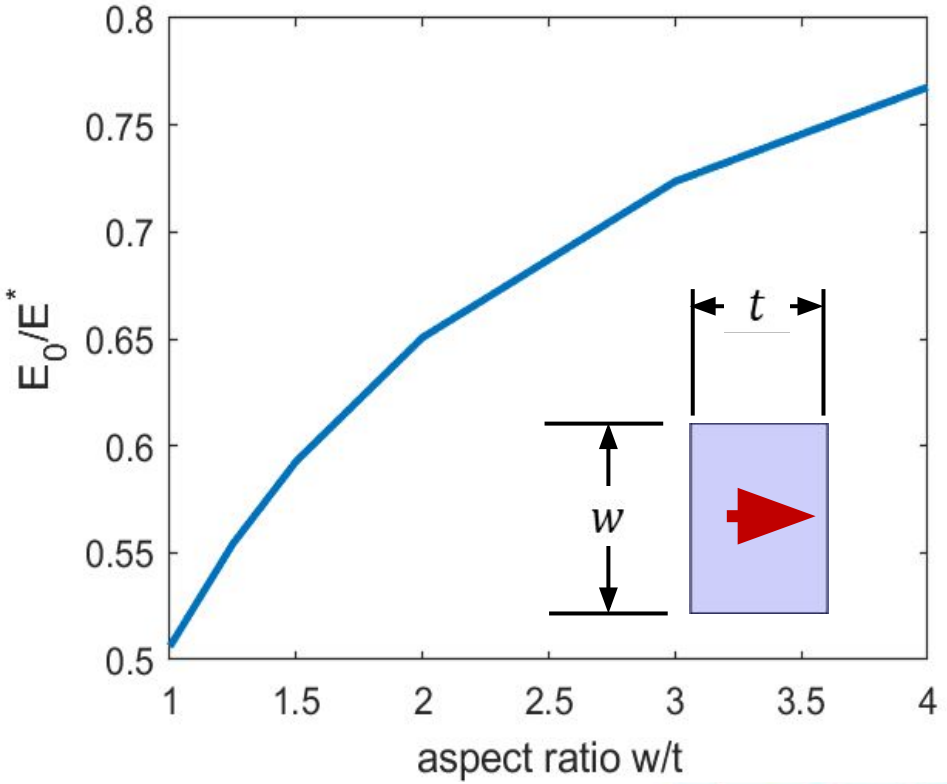
Permeability:
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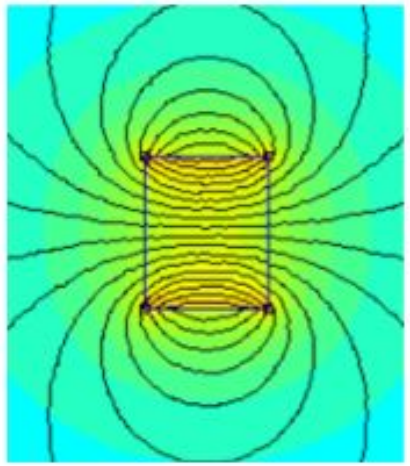
Magnetic energy and demagnetization curves

- B field "concentration" within a material is related to shape.
- $B_0 = N\, B_{ext}$



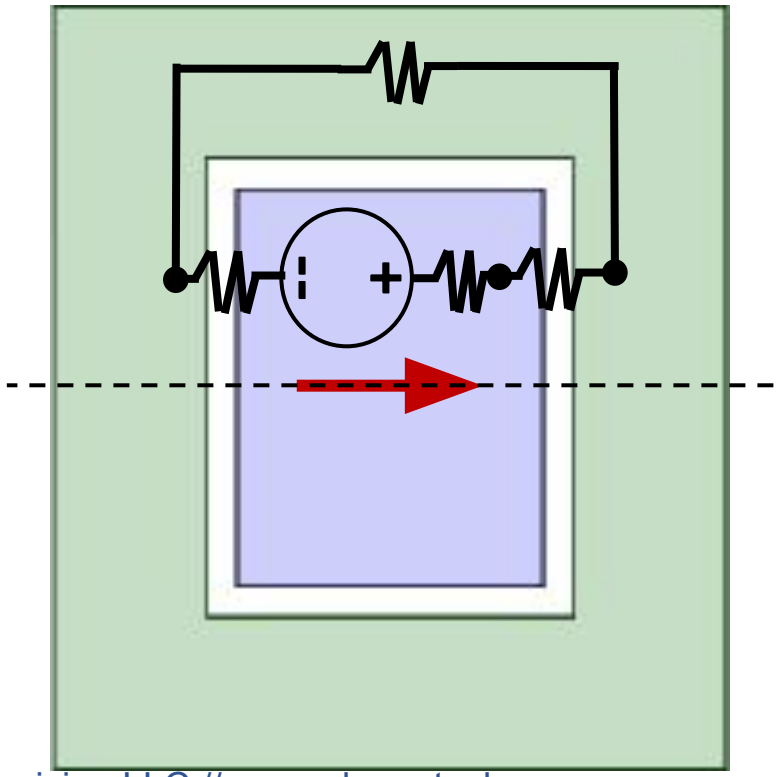
From:
www.mriquestions.com

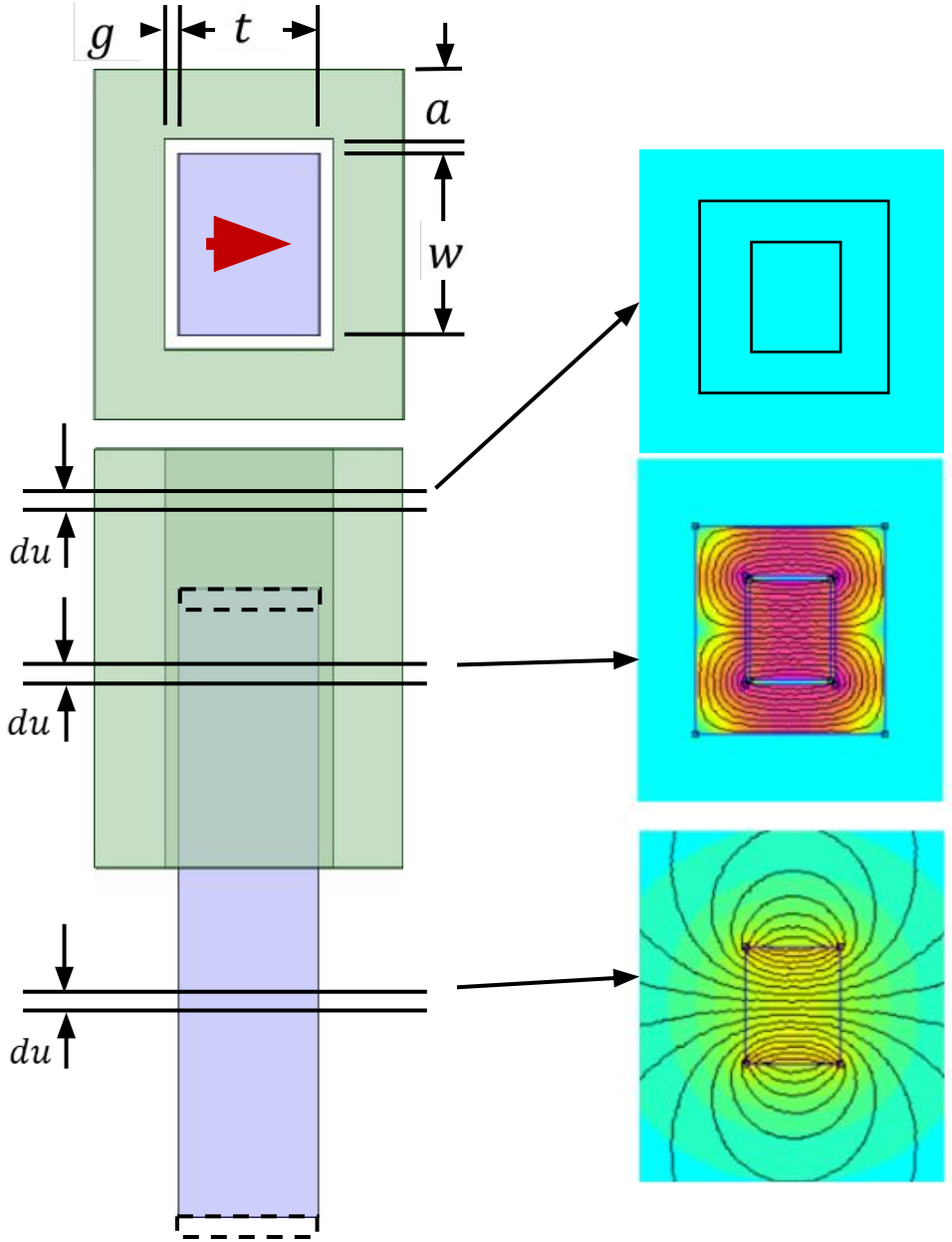




Shear configuration – magnet in an iron tube

- For a relatively long magnet and tube, the B field is nearly 2D with 3 distinct regions
- Magnetic circuit model of the magnet, air gaps, and iron.





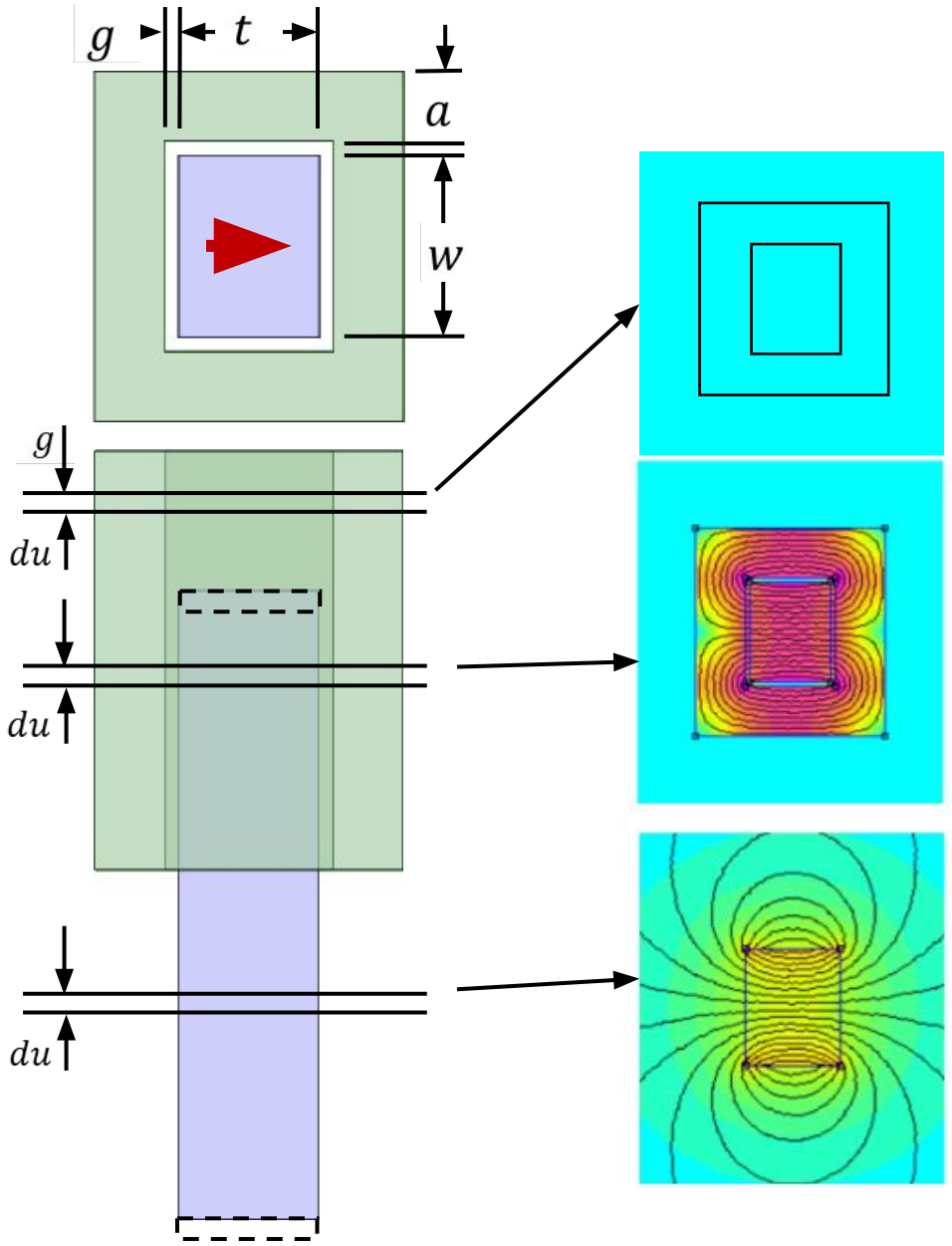
Shear configuration – magnet in an iron tube (cont)

- Define E as the energy per unit thickness of a slice
- Force is given by the change in energy.
  - $F \approx E_0 - E_1$
  - From the equivalent magnetic circuit, the B field in the air gap is

$$B = B_r \frac{t/\mu_r}{t/\mu_r + 2g}$$

  - For E_0, we can use a handbook formula or use 2D FEA
  - Nondimensionalize by the energy in a slice of magnet

$$E^* = \frac{B_r^2 t w}{2\mu_r \mu_0}$$



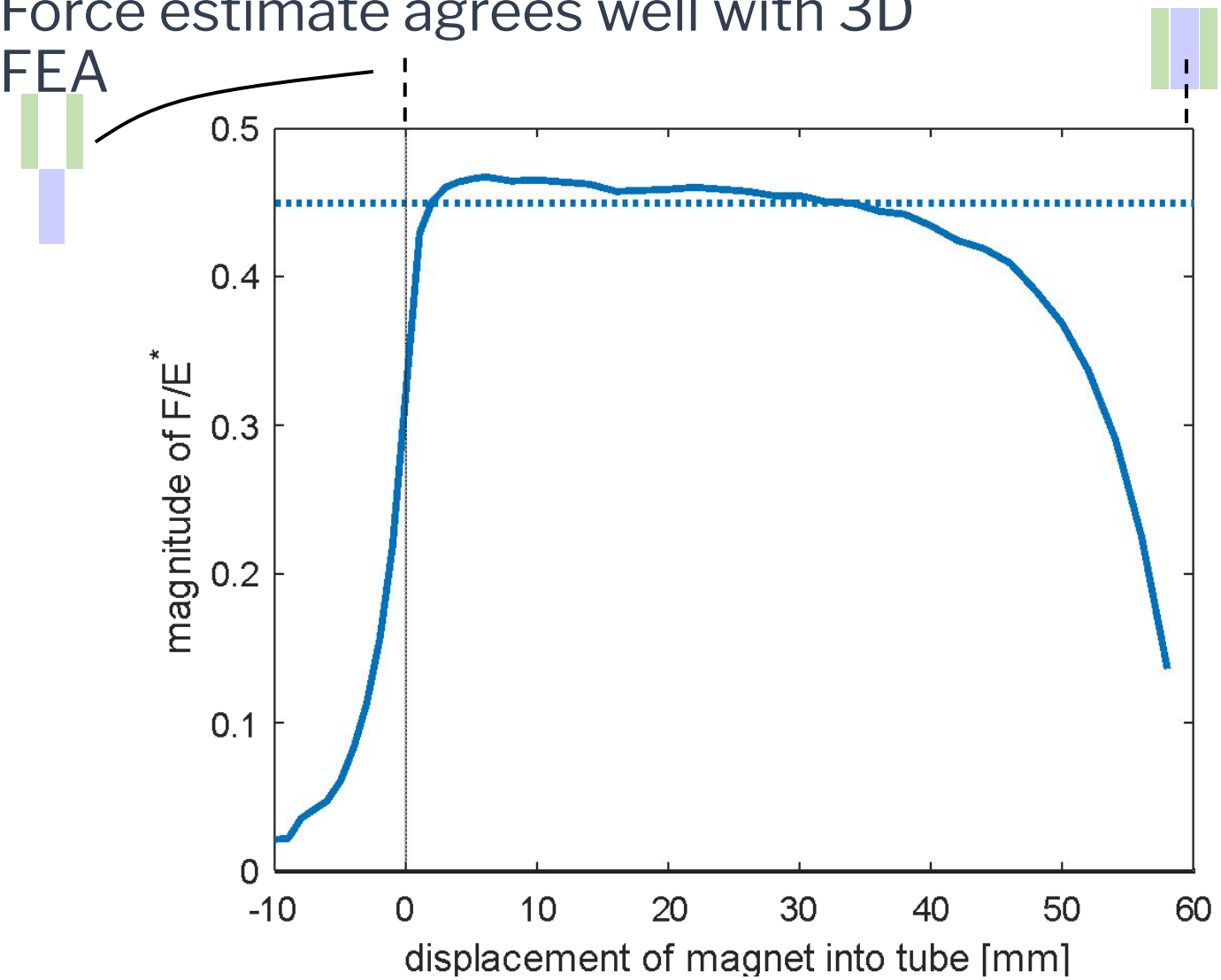
$E^* = 101$ J/m

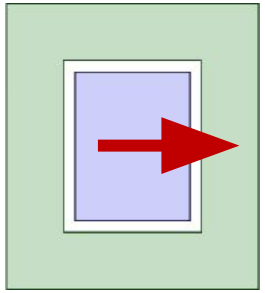
$E_1 = 0.09E^* = 9.1$ J/m

$E_0 = 0.55E^* = 56$ J/m

Shear configuration – magnet in an iron tube

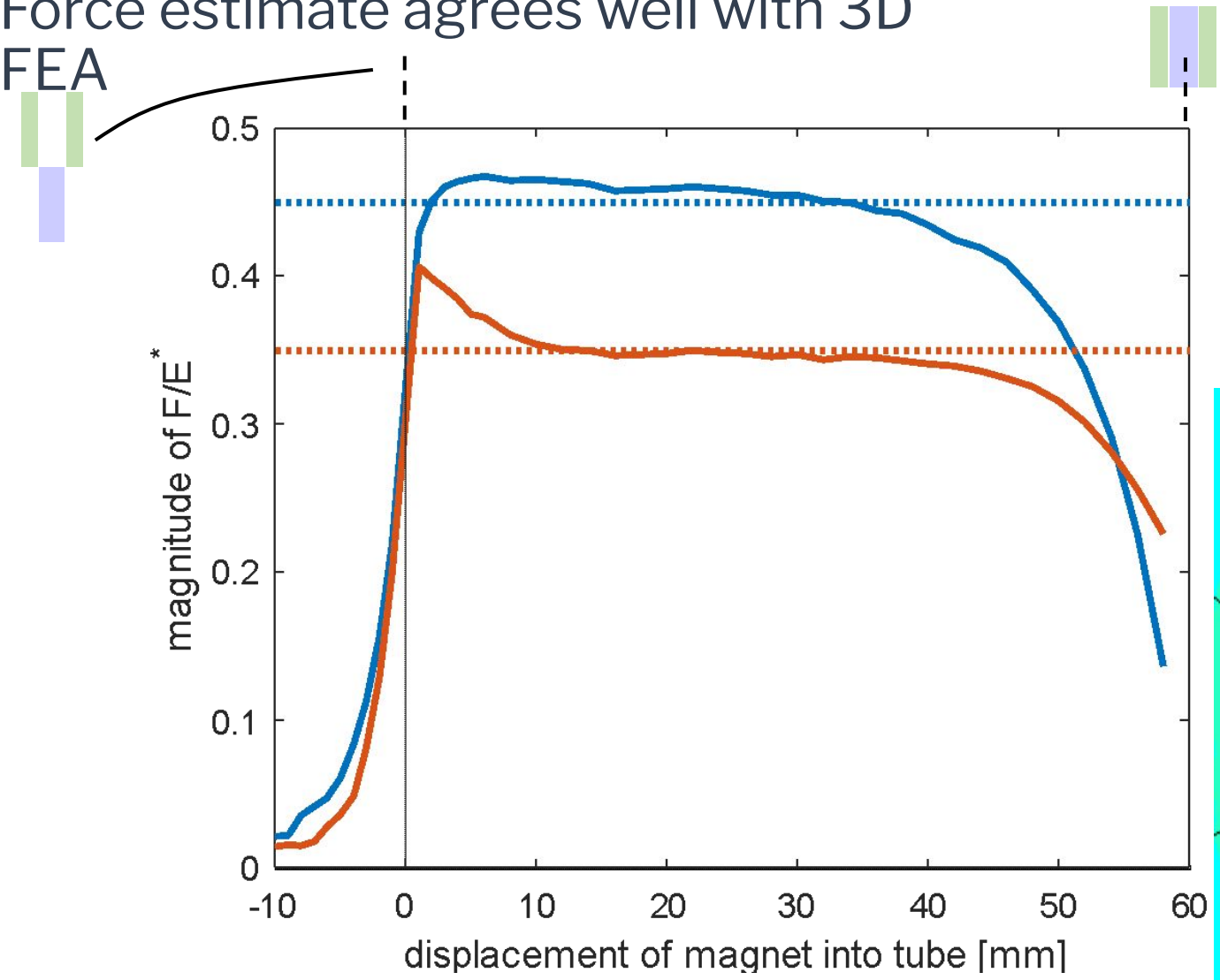
- Force estimate agrees well with 3D FEA



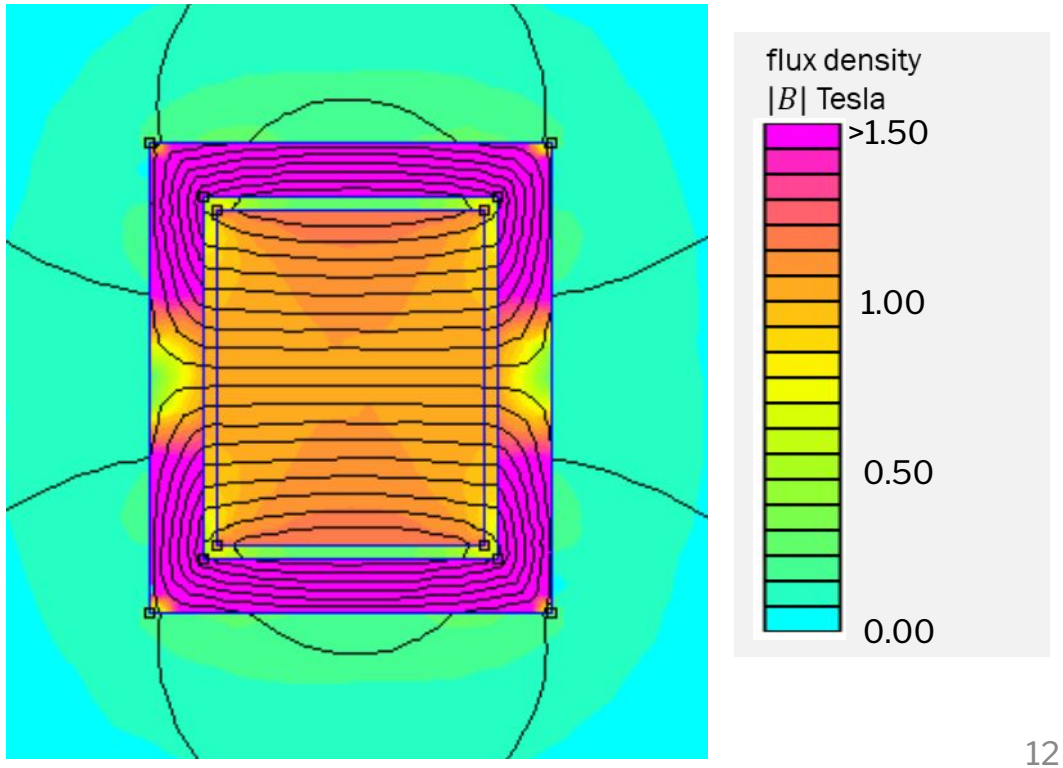


Shear configuration – magnet in an iron tube

- Force estimate agrees well with 3D FEA

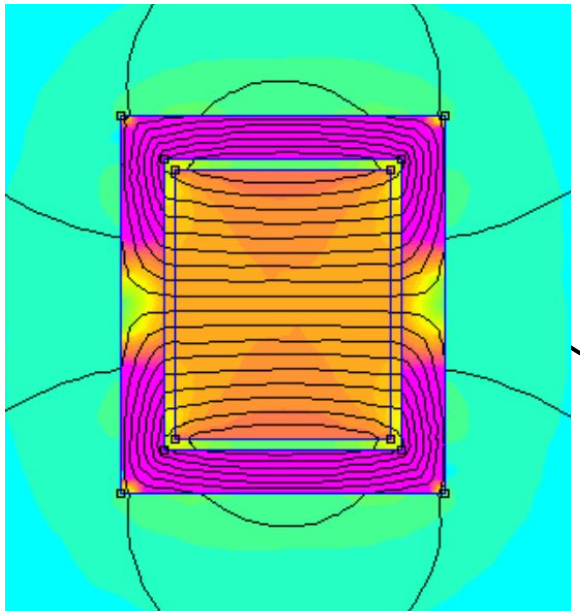


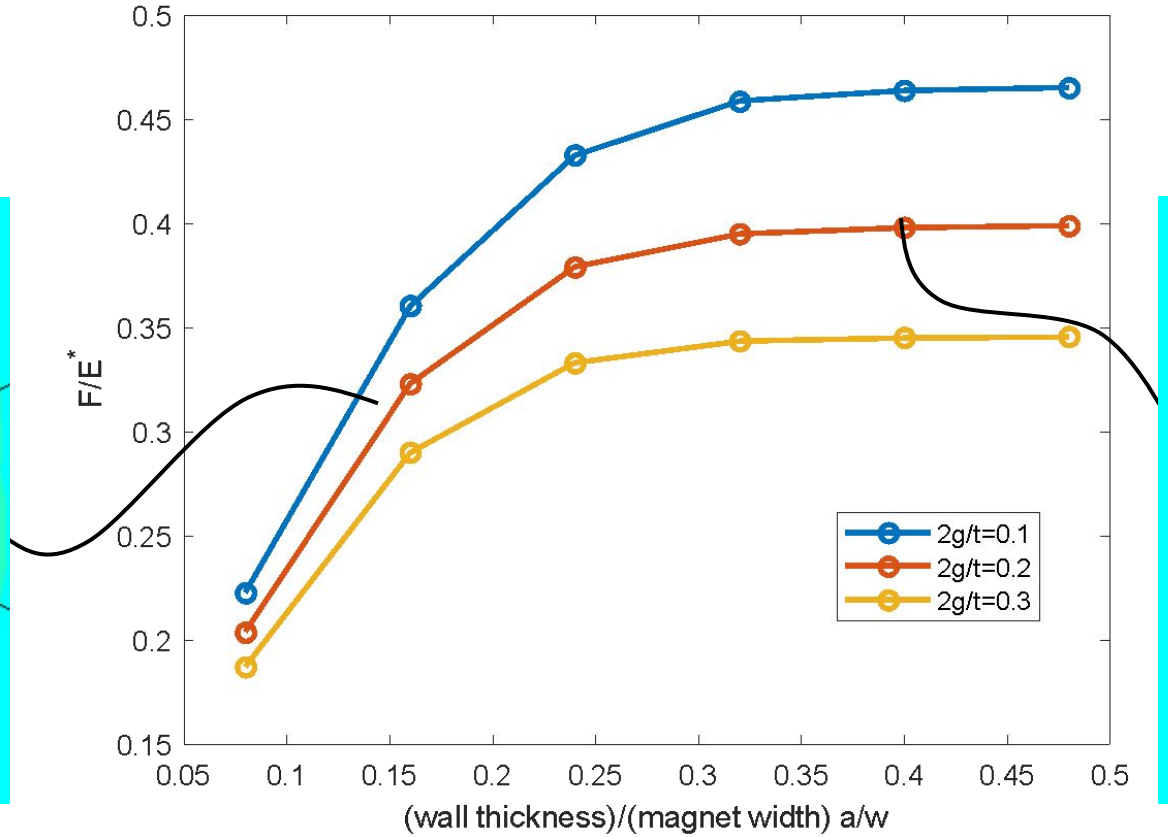
- If the iron is thinned, saturation reduces the force.
  - This can be seen using the same method but E_1 comes from 2D FEA not analytical formula.

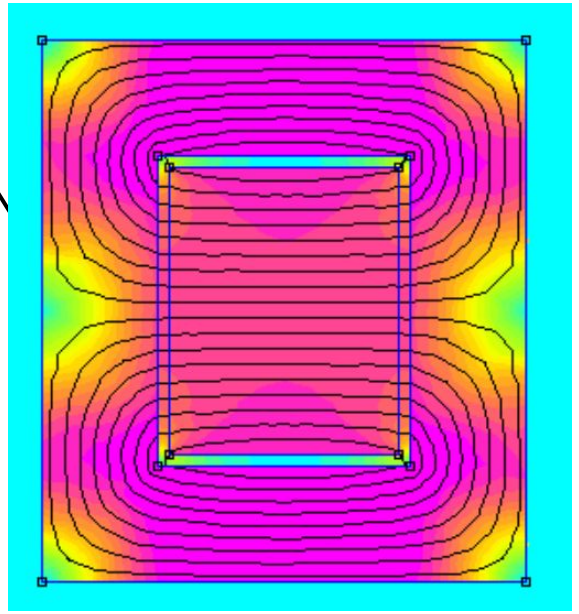


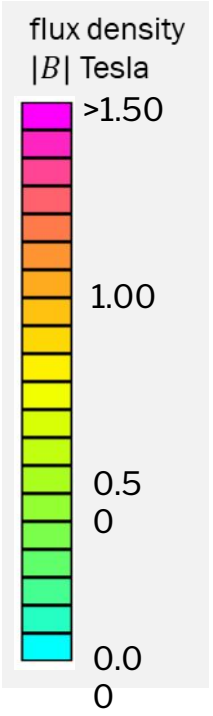
Sensitivity to Air Gap and Wall Thickness

- For N52 magnets and 1018 steel, wall thickness about 1/3 magnet width suffices.



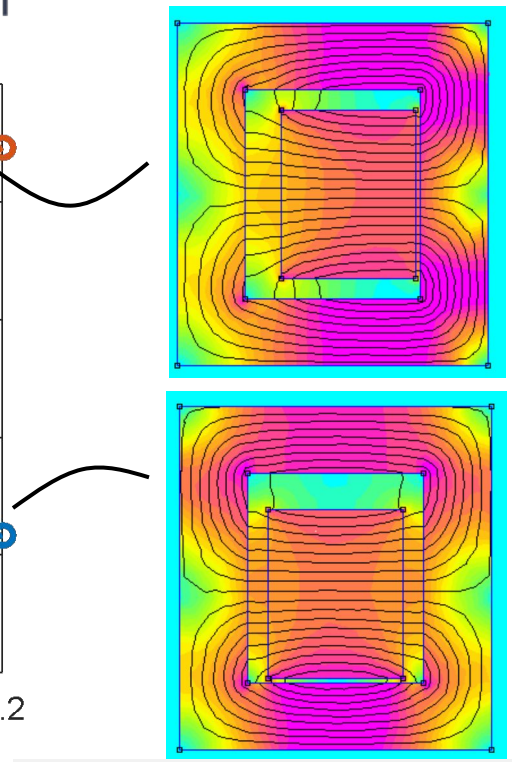


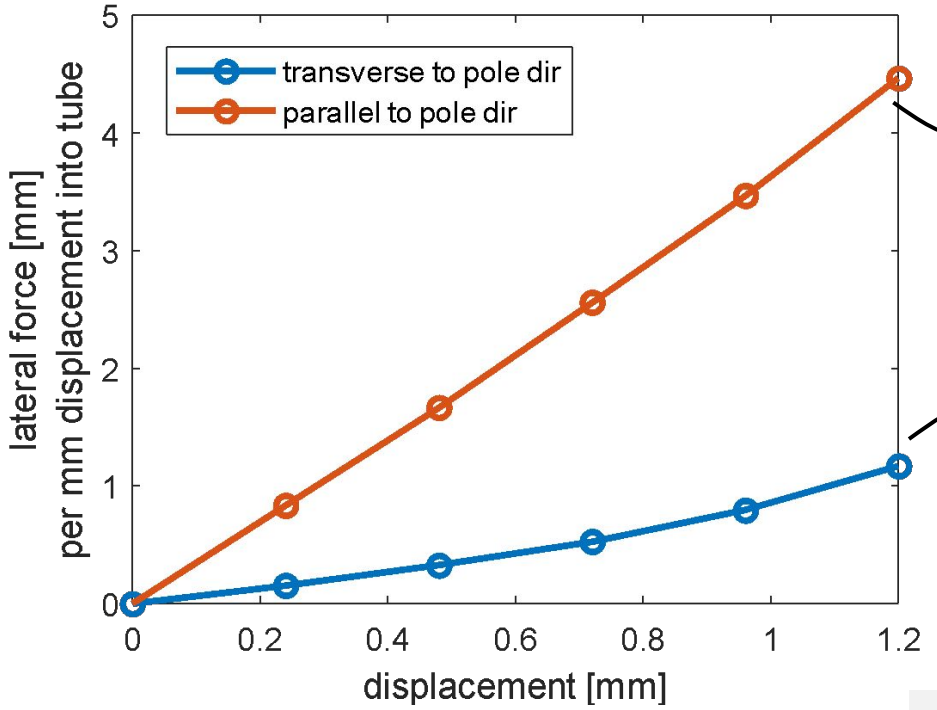


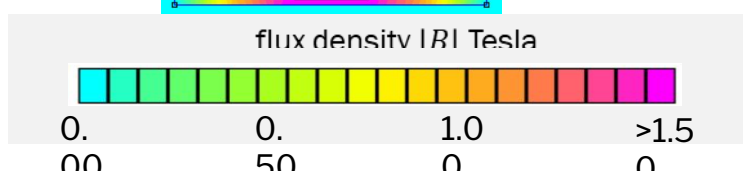


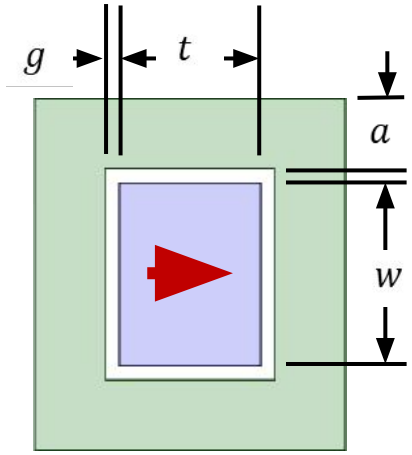
Lateral Negative Stiffness

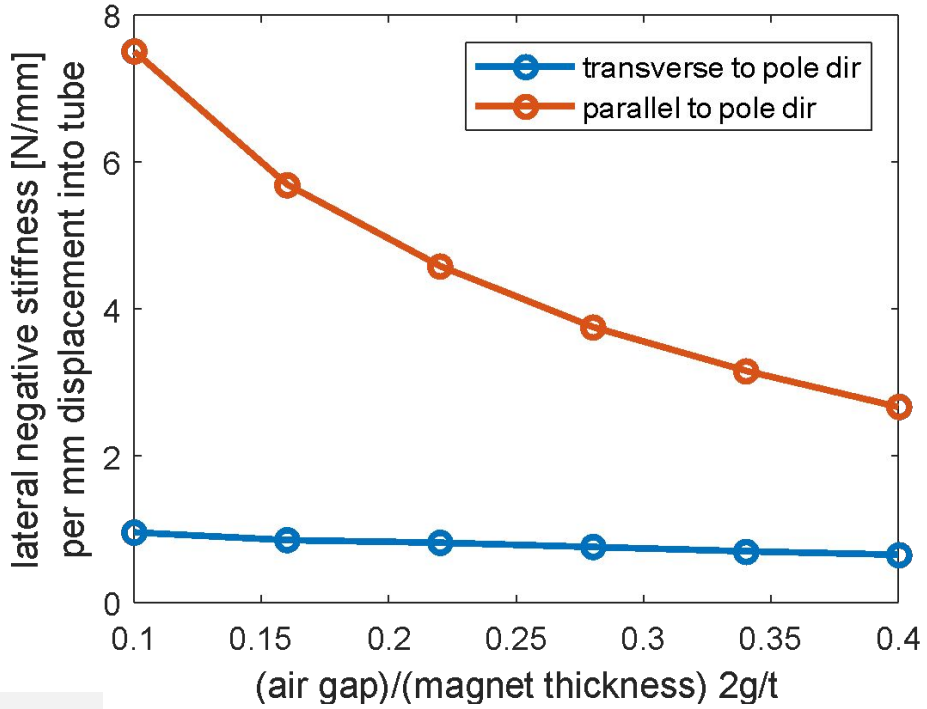
- When the magnet is displaced laterally, it is attracted to the tube wall.
- The resulting negative stiffness is much larger in the pole direction than in the transverse direction.
- Results for an N52 magnet in 1018 tube with
  - $w = 12, t = 10, a = 6$ mm





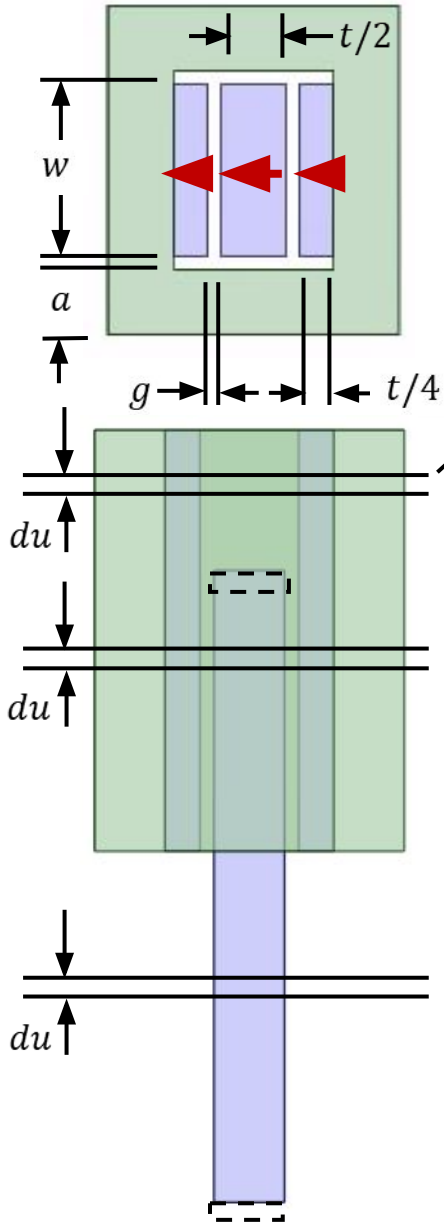


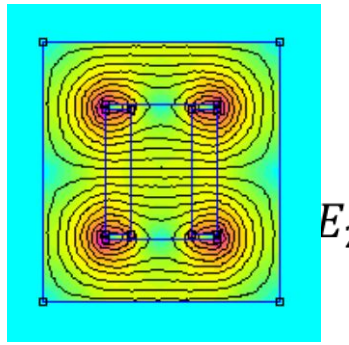




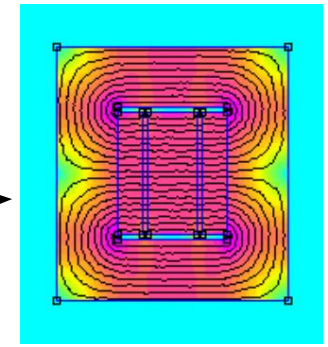
Split magnet configuration of magnet in a tube

- By dividing the magnet into separate pieces, the moving mass is reduced.
- Force is the same
- $F = (E0 - (E1 - E2))$
- $F = 50$ N

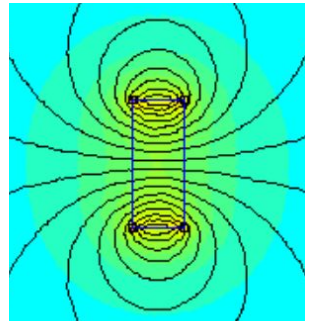




$E_2 = 0.24E^* = 24$ J/m



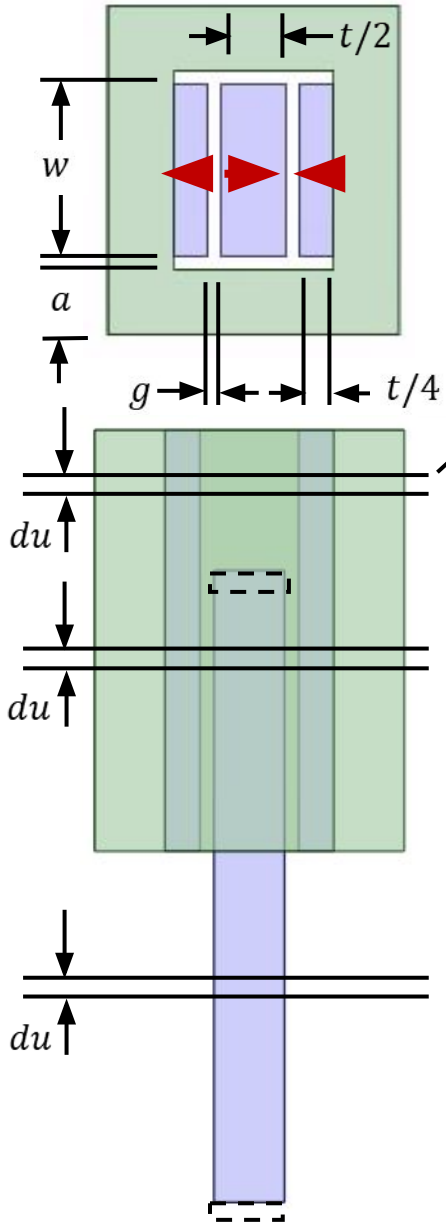
$E_1 = 0.095E^* = 9.6$ J/m

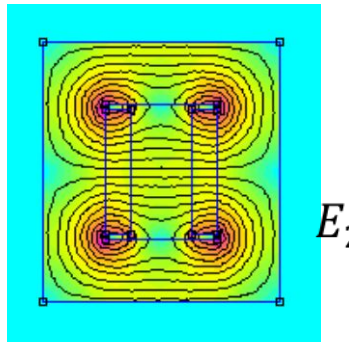


$E_0 = 0.35E^* = 35$ J/m

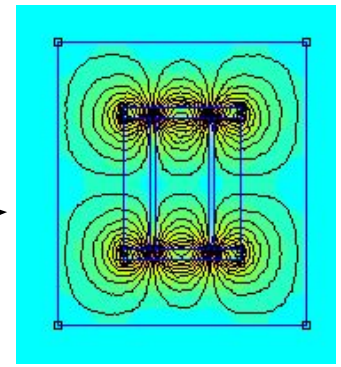
Repelling configuration of magnet in a tube

- By reversing the middle magnet, the force is now repulsive.
- The magnitude is about half
- $F = (E0 - (E1 - E2))$
- $F = -23$ N

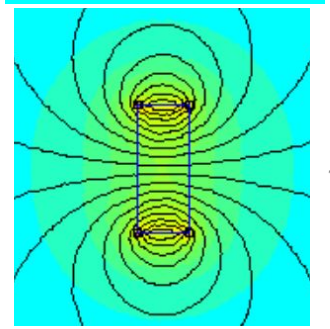




$E_2 = 0.24E^* = 24$ J/m



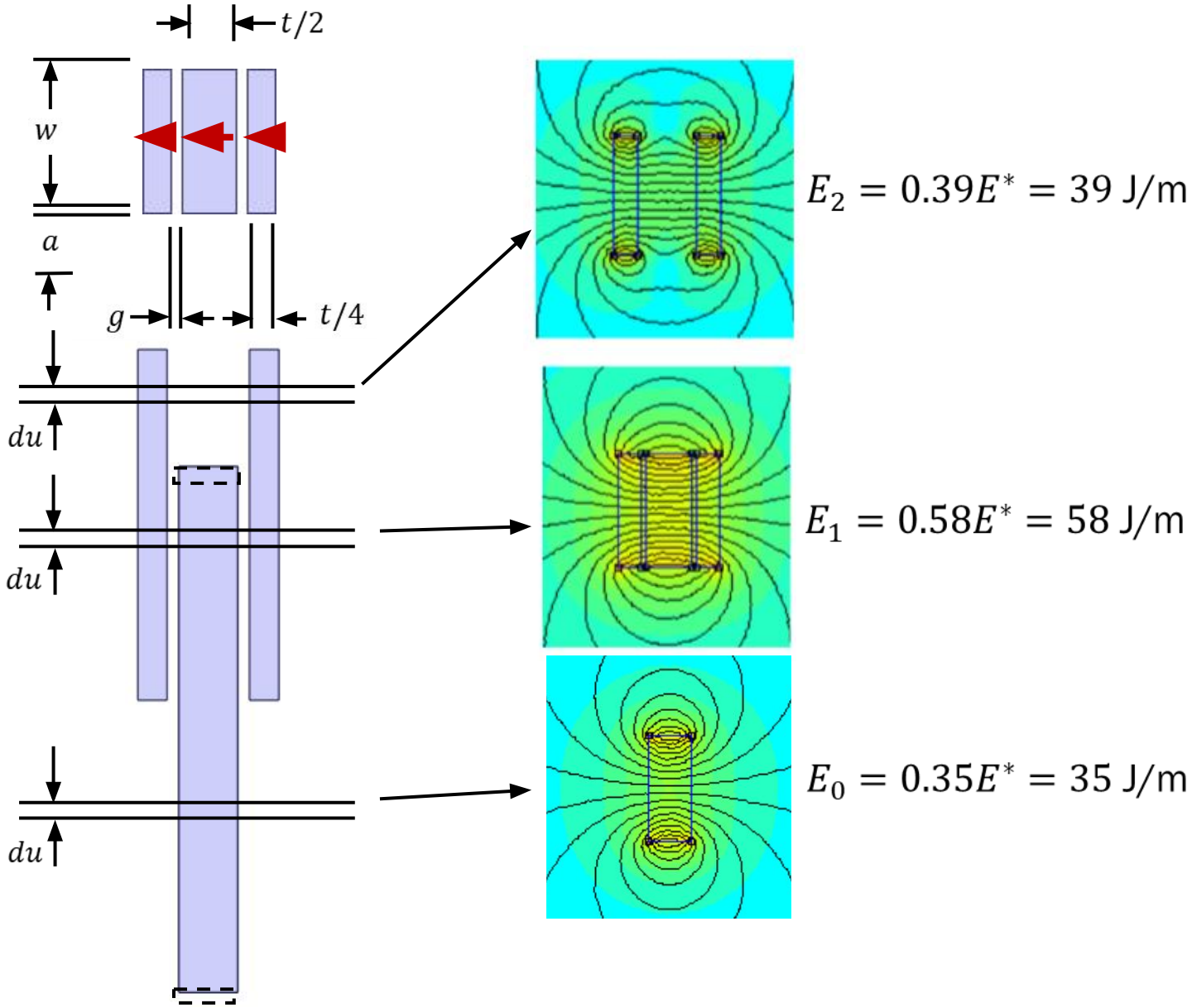
$E_1 = 0.83E^* = 83$ J/m



$E_0 = 0.35E^* = 35$ J/m

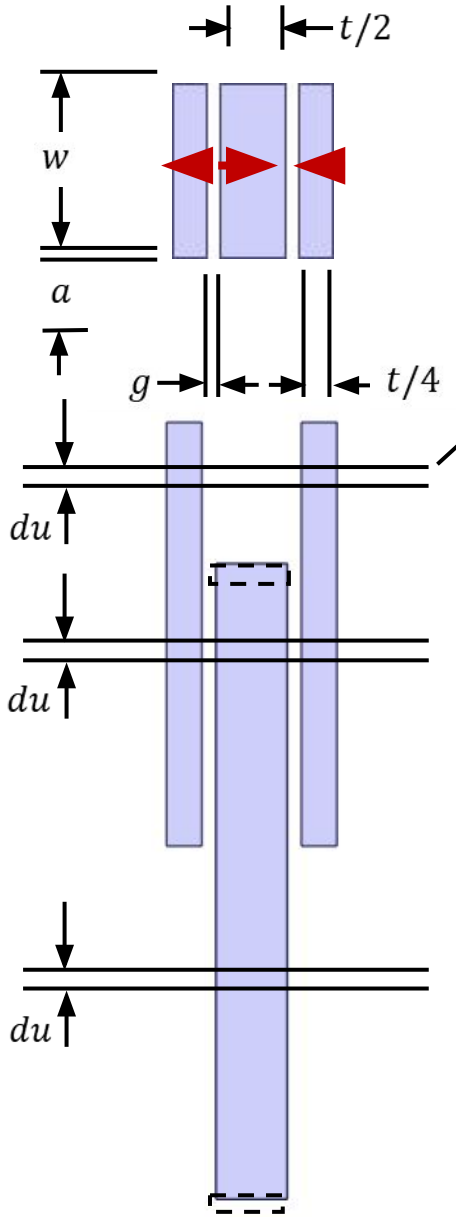
Ironless shear configuration

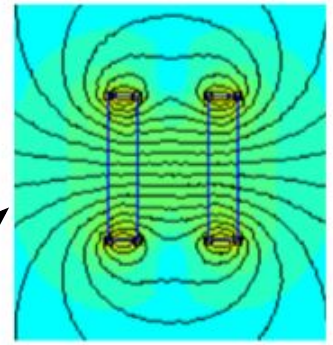
- Removing the iron eliminates a large source of hysteresis
- The magnitude is about 1/3 compared to with iron.
- But if the magnets are increased in size to occupy the same total size as the iron case, then the force is about the same.
- $F = (E0 - (E1 - E2))$
- $F = 16$ N



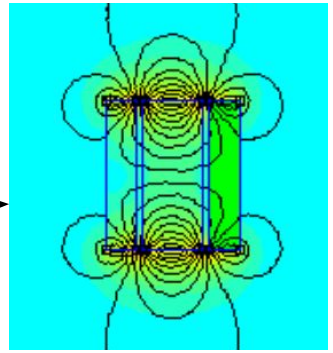
Ironless shear repulsive configuration

- If the middle magnet is reverse, the force is repulsive.
- The force is only slightly reduced from the attracting case.
- $F = (E0 - (E1 - E2))$
- $F = -14$ N

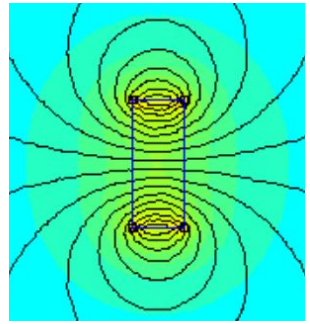




$E_2 = 0.39E^* = 39$ J/m

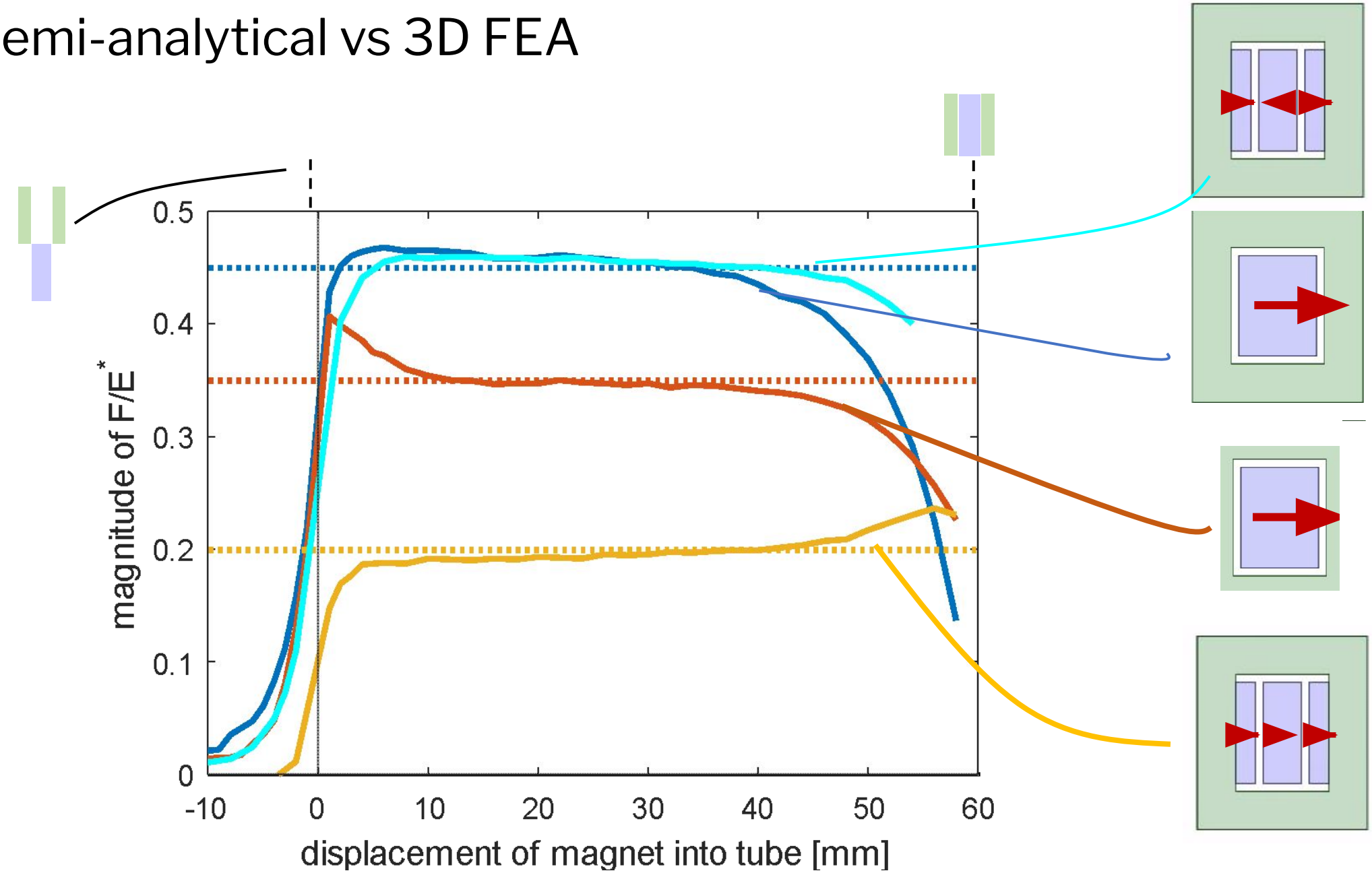


$E_1 = 0.88E^* = 88$ J/m



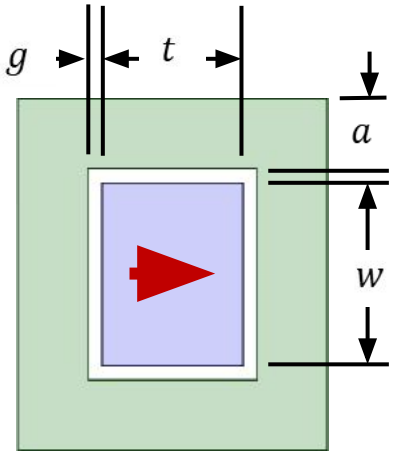
$E_0 = 0.35E^* = 35$ J/m

Semi-analytical vs 3D FEA

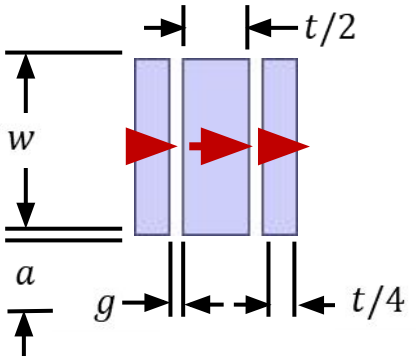


Comparison of Force with and Without Soft Magnetic Tube

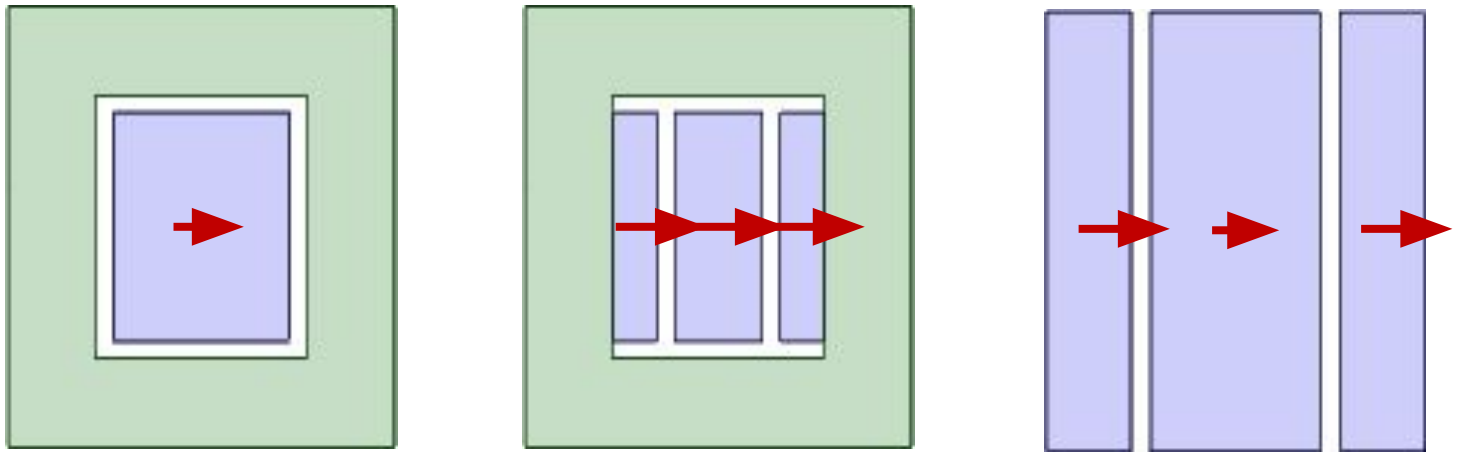
- Set gap to $1/10^{\text{th}}$ the thickness so that $2g/t = 0.1$
- With tube thickness $a = w/3$, we attain $F \approx 0.45E^*$.
  - So the overall width of the tube is about $1.7w$.
- If we use the three magnet without tube configuration with the same air gap, we attain $0.15E^*$.
- E^* is based on the total magnet area, to get the same force without the tube we need three times the area, or approximately 1.7 times the linear dimension.
- Drawn to scale, the three figures below yield approximately the same force.



$F \approx 0.45E^*$

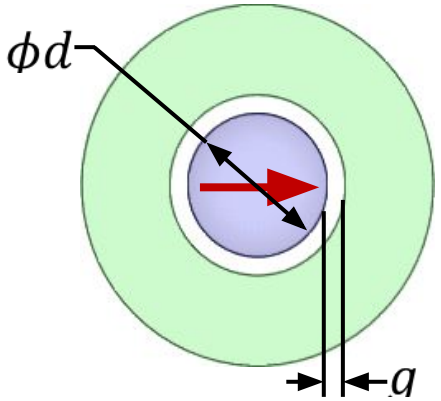


$F \approx 0.15E^*$

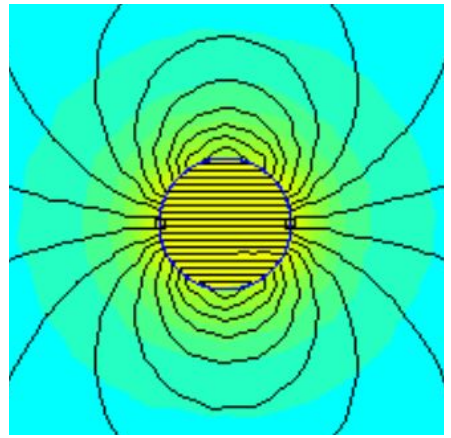


Diametrically-Poled Magnet in a Circular Tube

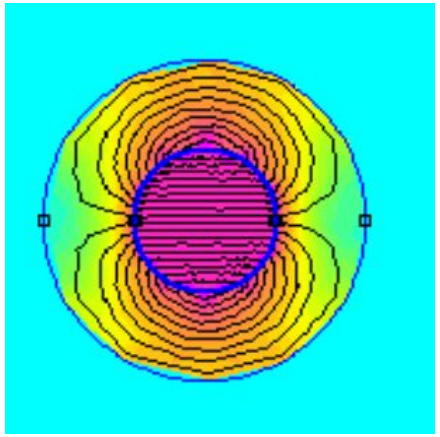
- For a circular magnet of diameter d
  - $E^* = \pi d^2 B_r^2 / 8\mu_r\mu_0$
  - The demagnetization is $E_0/E^* = 1/2$
- We can obtain E_1/E^* from 2D FEA
- Then the force is $F = E_0 - E_1$

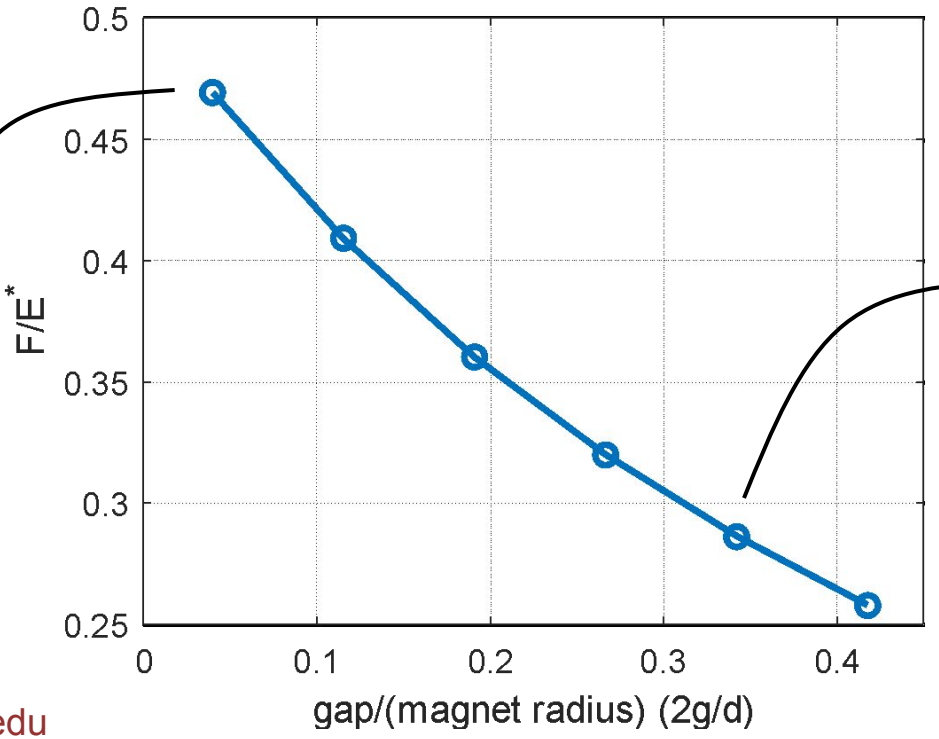


computation of E_0

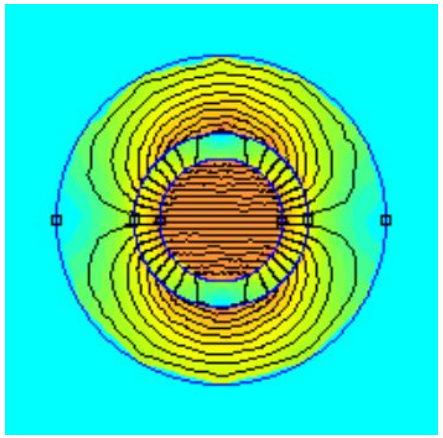


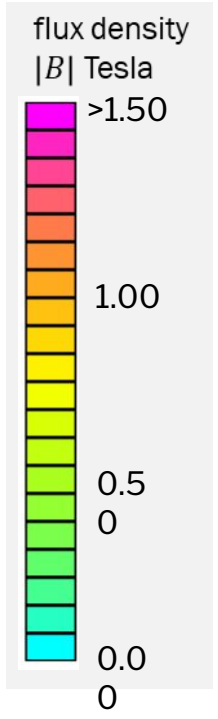
computation of E_1 for $2g/d = 0.03$





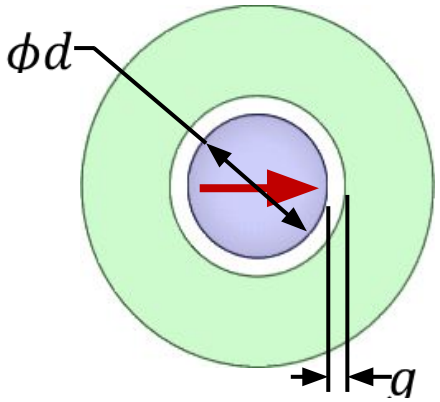
computation of E_1 for $2g/d = 0.3$

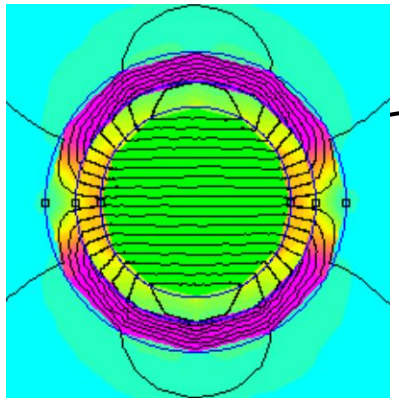


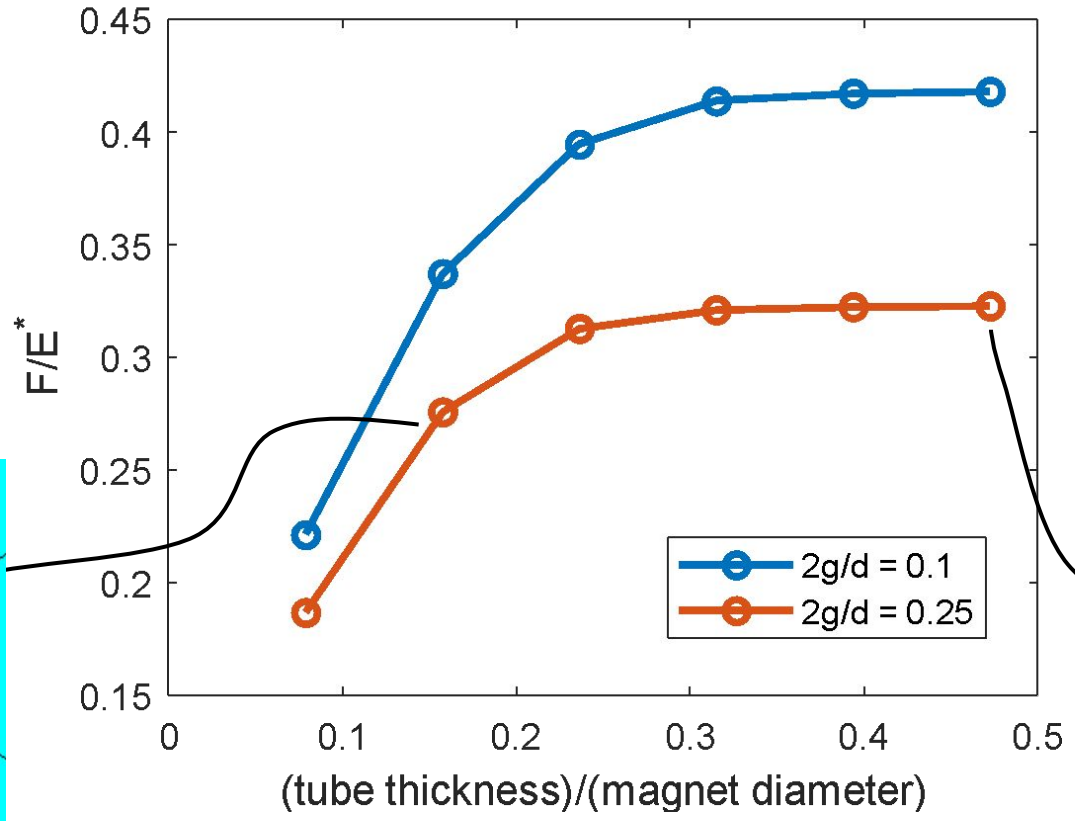


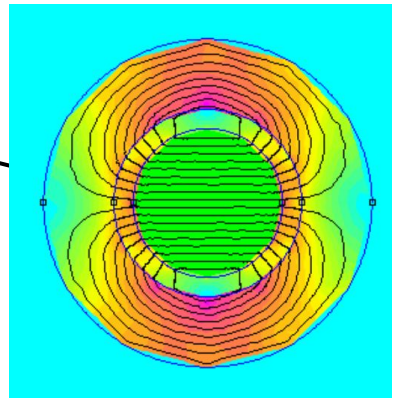
How thick Should the Tube be Made?

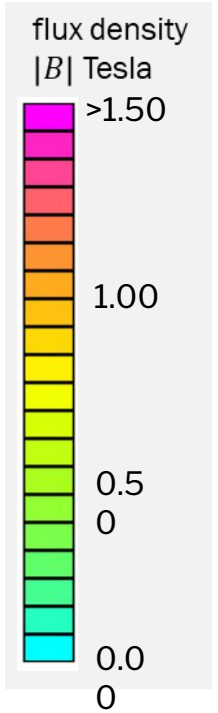
- For N52 magnets and 1018 steel, tube thickness about 1/3 magnet diameter suffices.





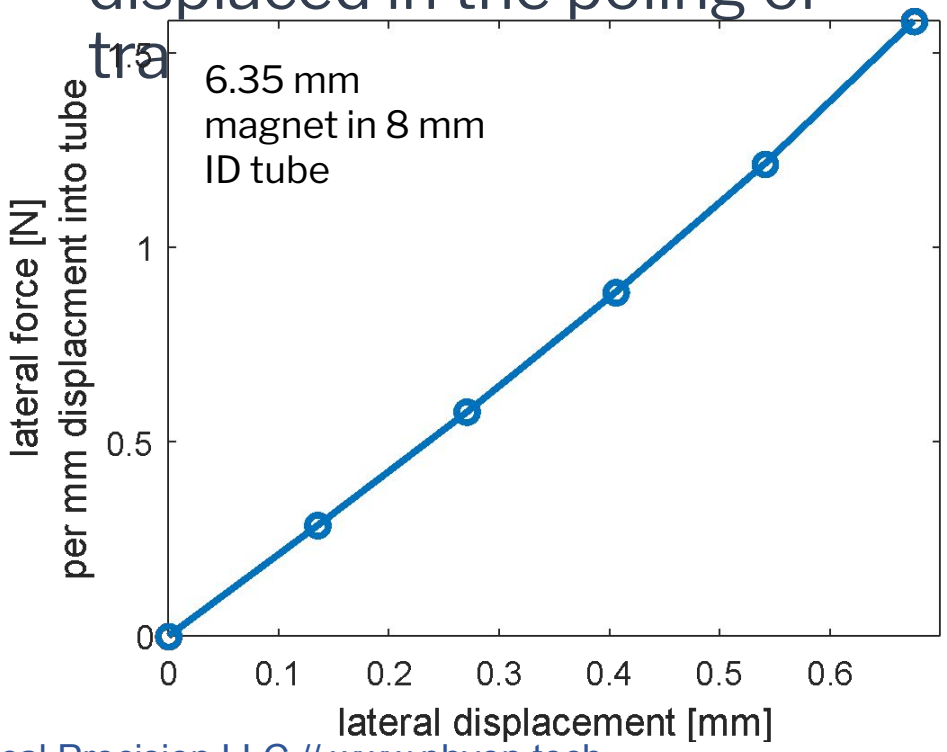


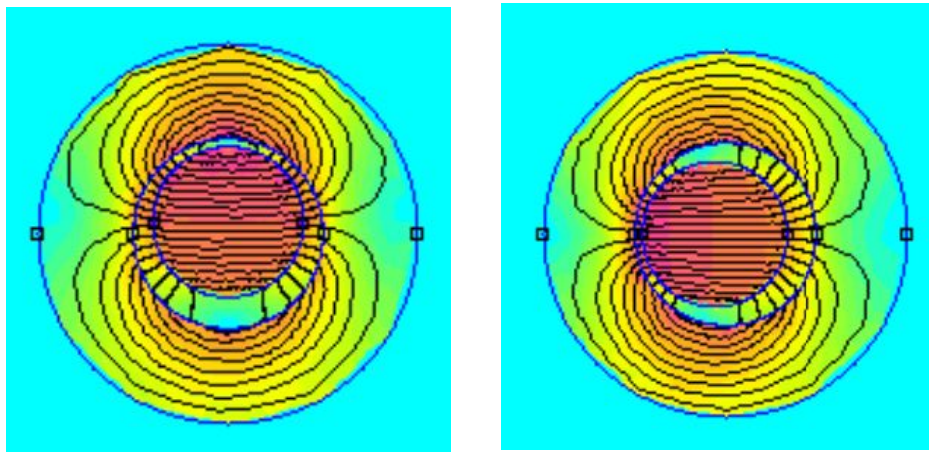


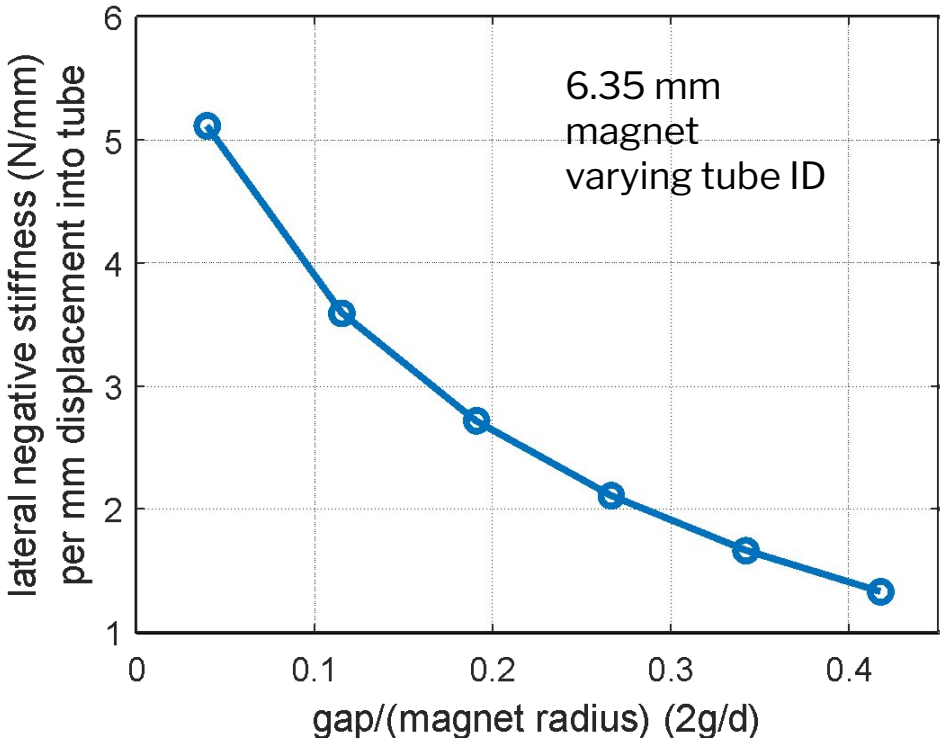


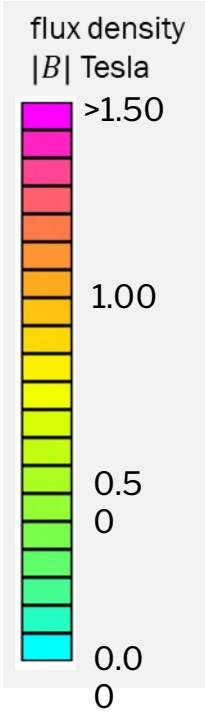
Lateral Negative Stiffness

- If the magnet is displaced laterally, it becomes attracted to the tube wall.
- It turns out that the attraction is the same whether the tube is displaced in the poling or tra



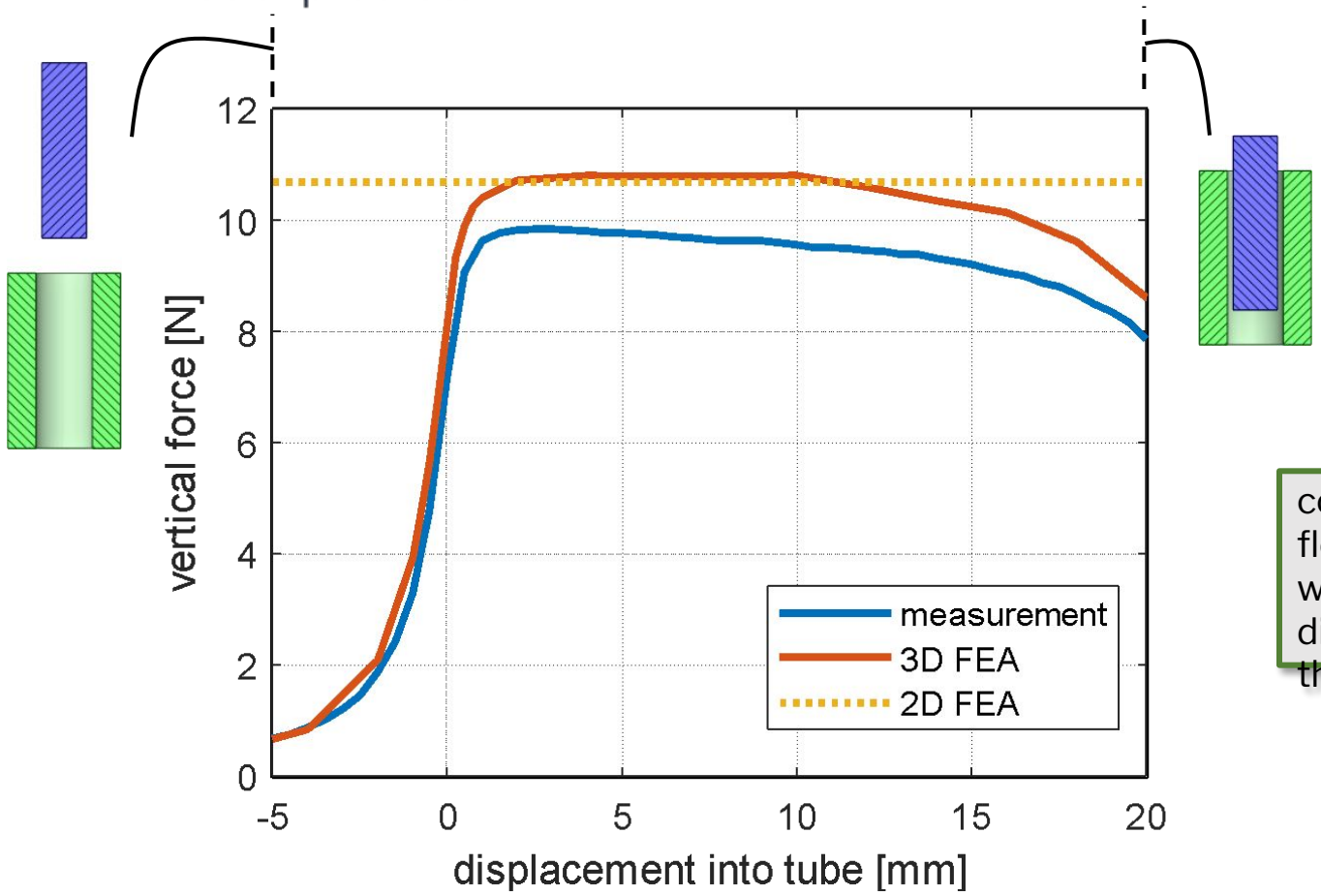


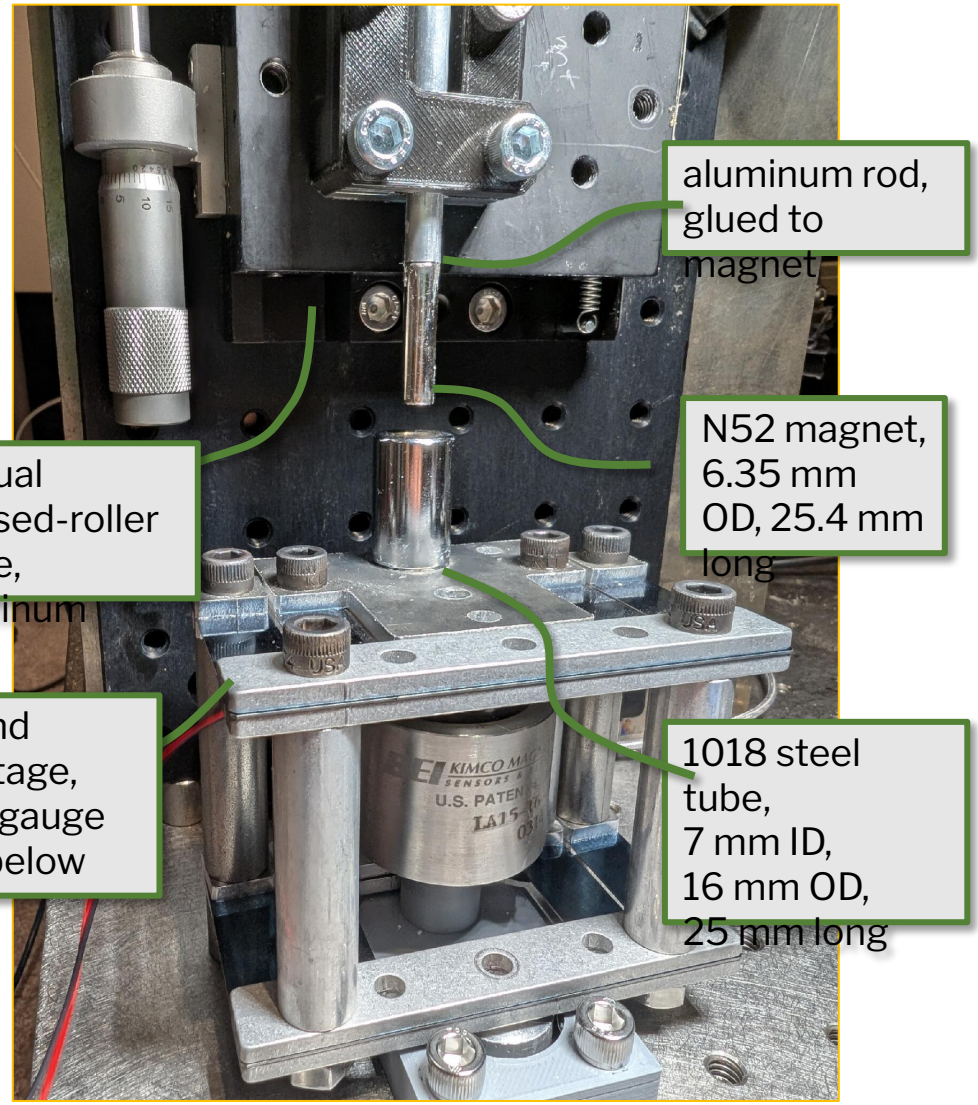




Experiment

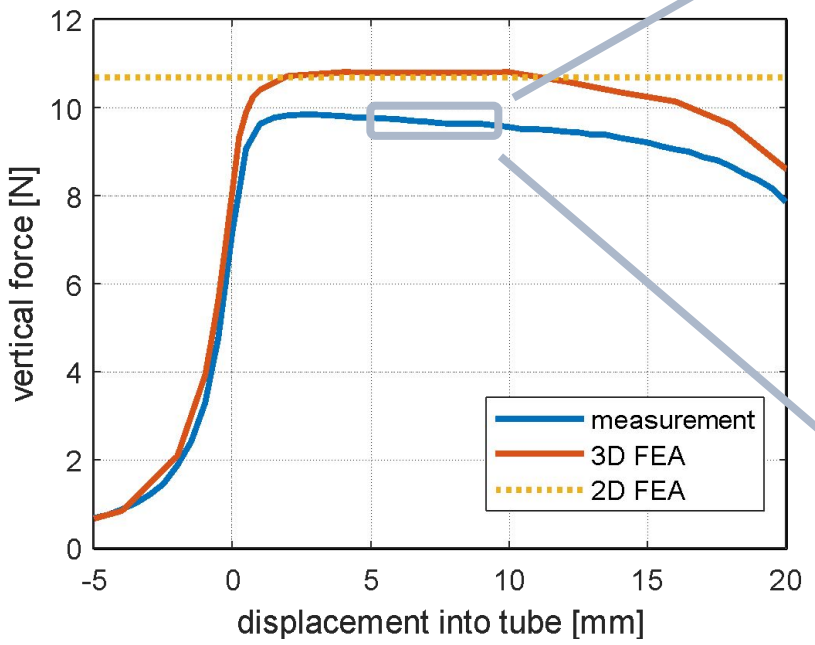
- $E^* \approx 26$ J/m for ¼ inch N52 magnet
- For a 7 mm ID tube, gap/magnet radius is about 0.1
- 2D model predicts $F \approx 0.42E^* \approx 11$ N

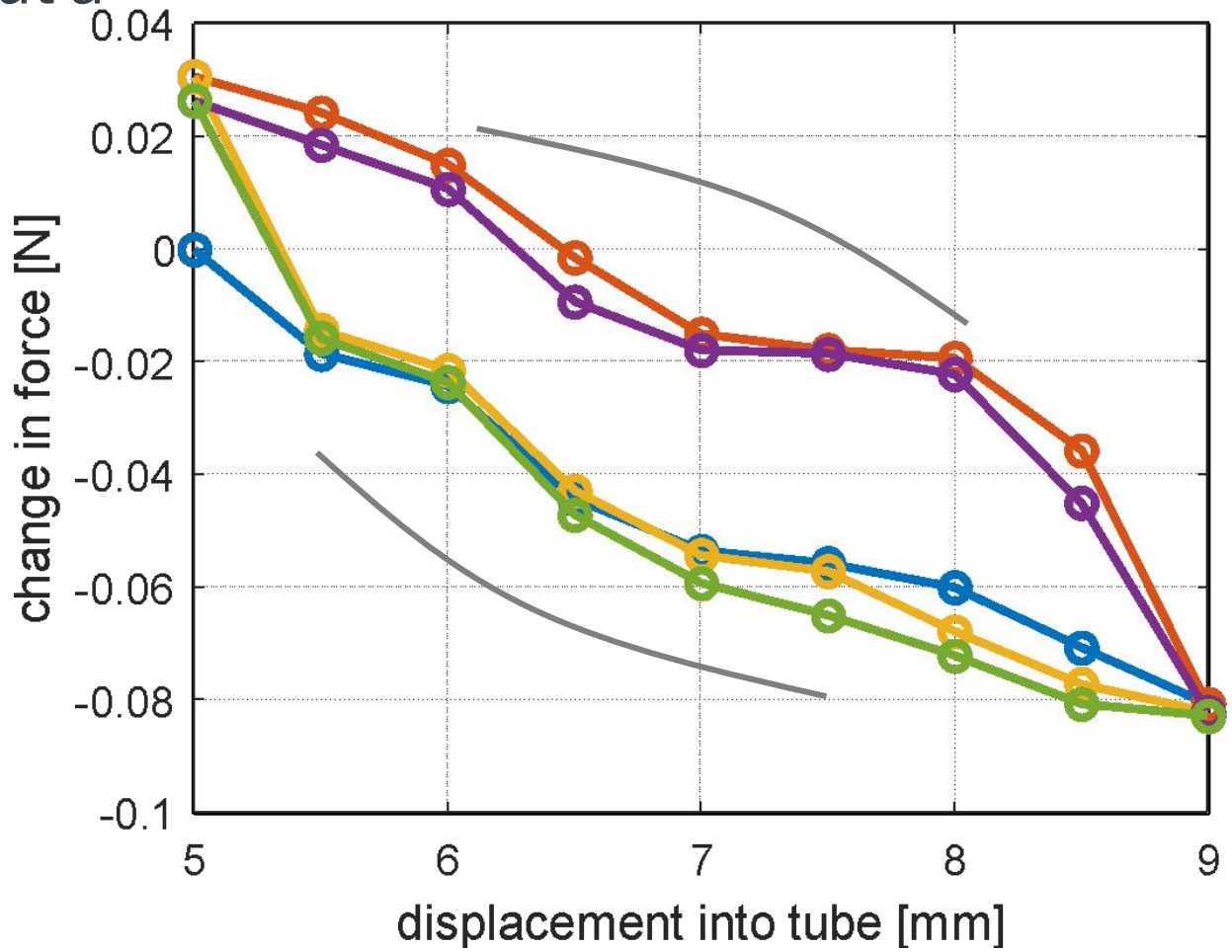




Hysteresis

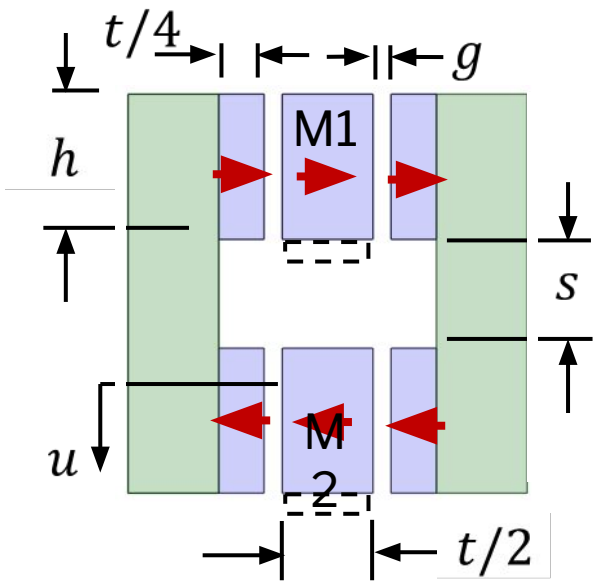
- The iron tube exhibits some hysteresis.
- By cycling the magnet back and forth over several mm, we measure about a 60 mN force loop, less than 1%

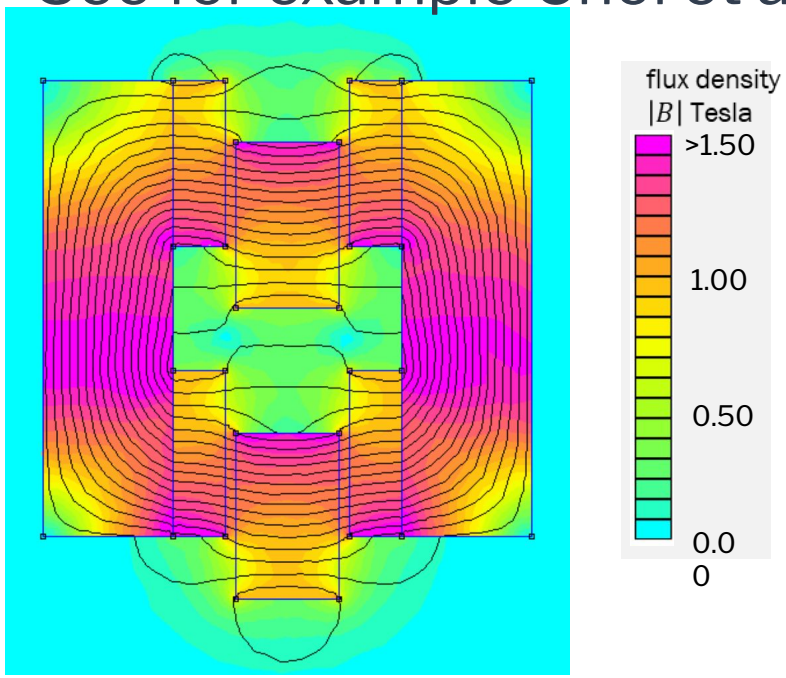


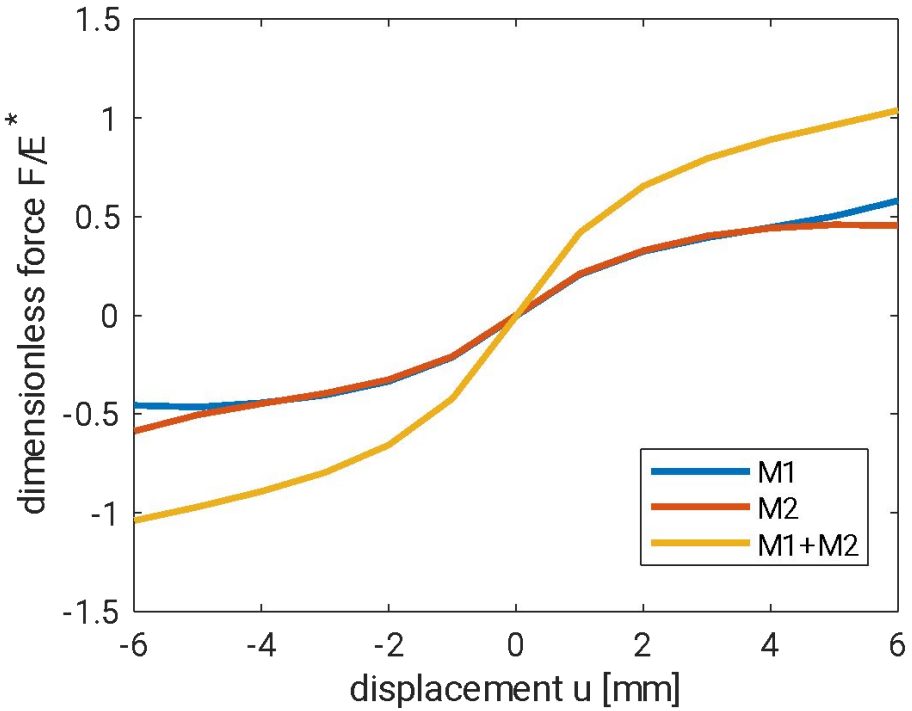


Shear configuration for moderate travel ranges

- For a single magnet, the force is nearly constant over a moderate travel range.
- Can package easily with a coil to also be an actuator.
  - See for example Choi et al 2009

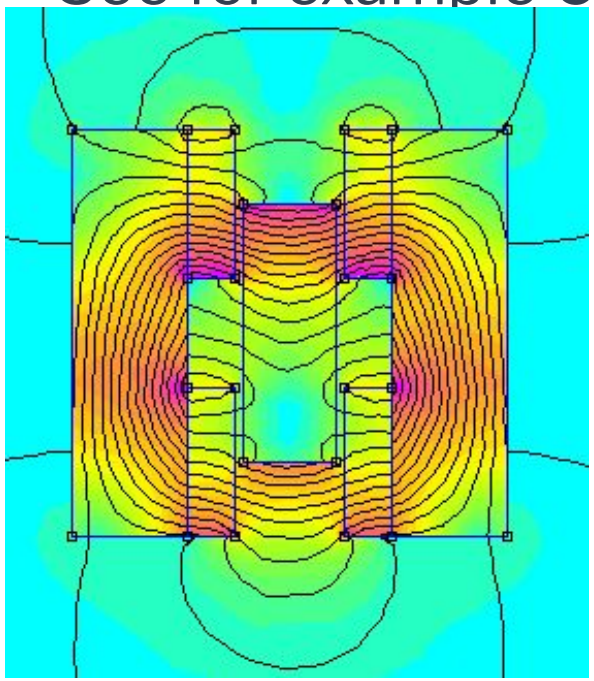


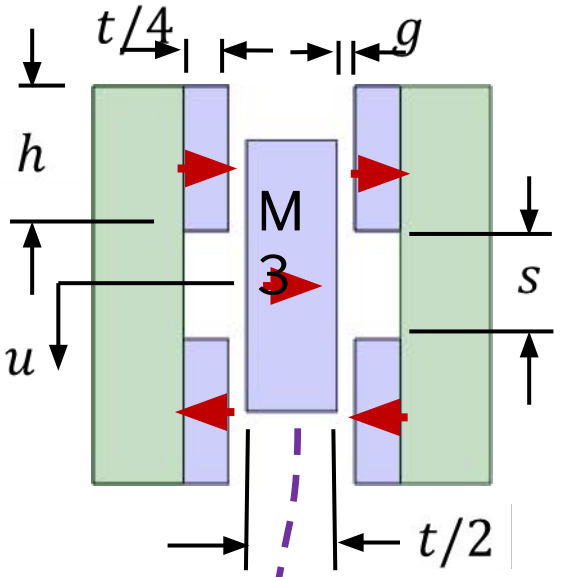


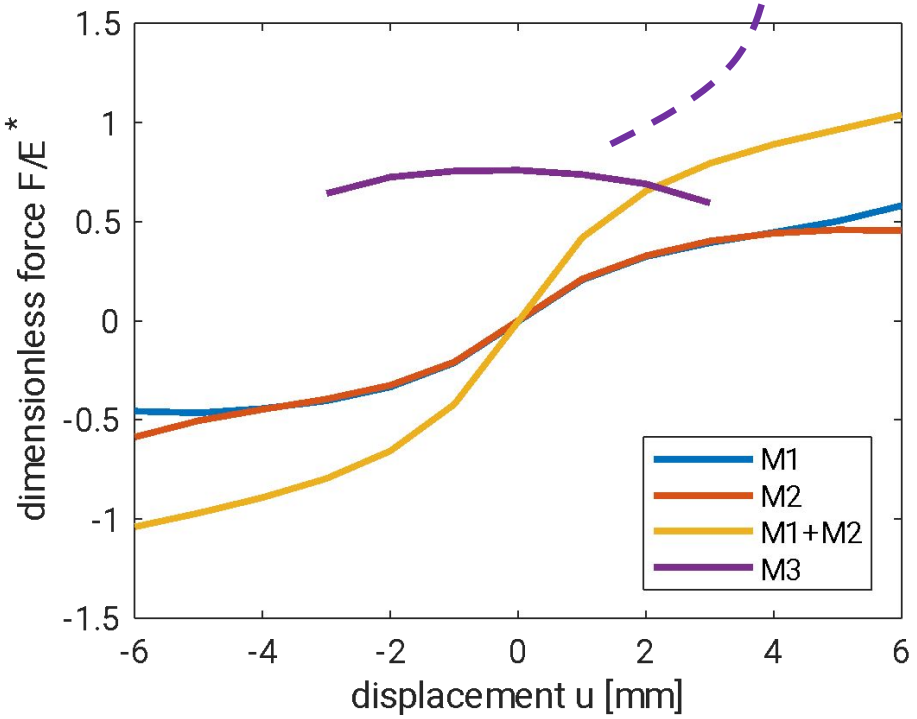


Shear configuration for moderate travel ranges

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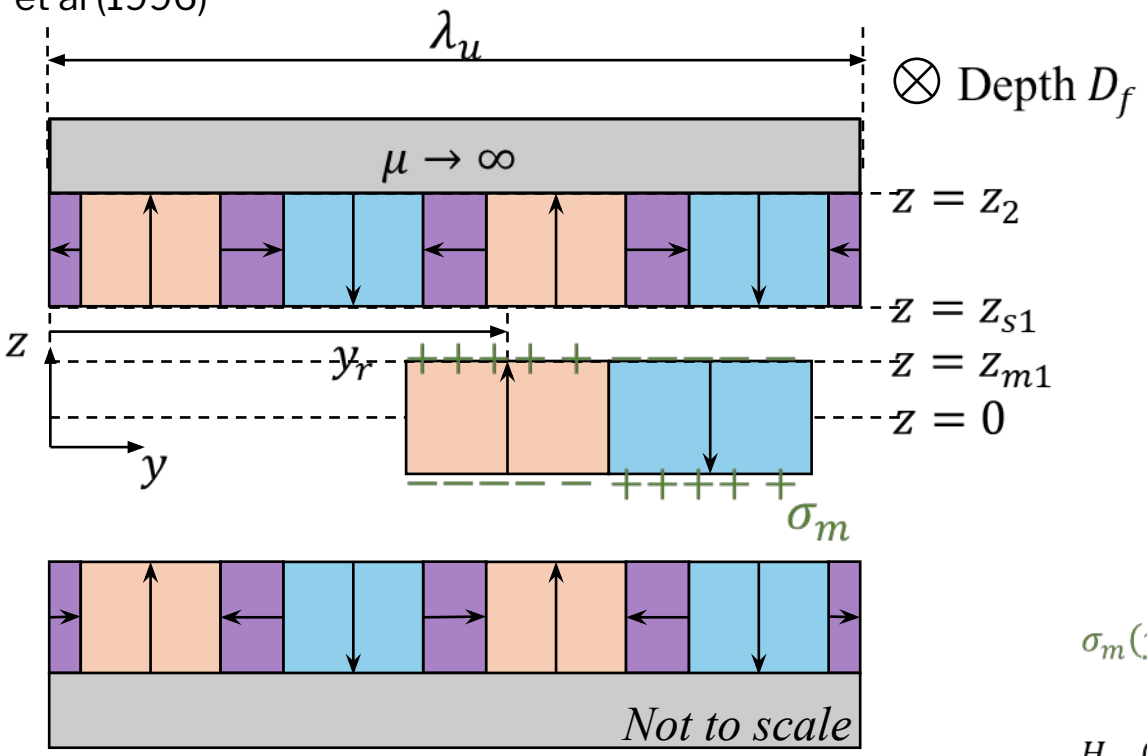


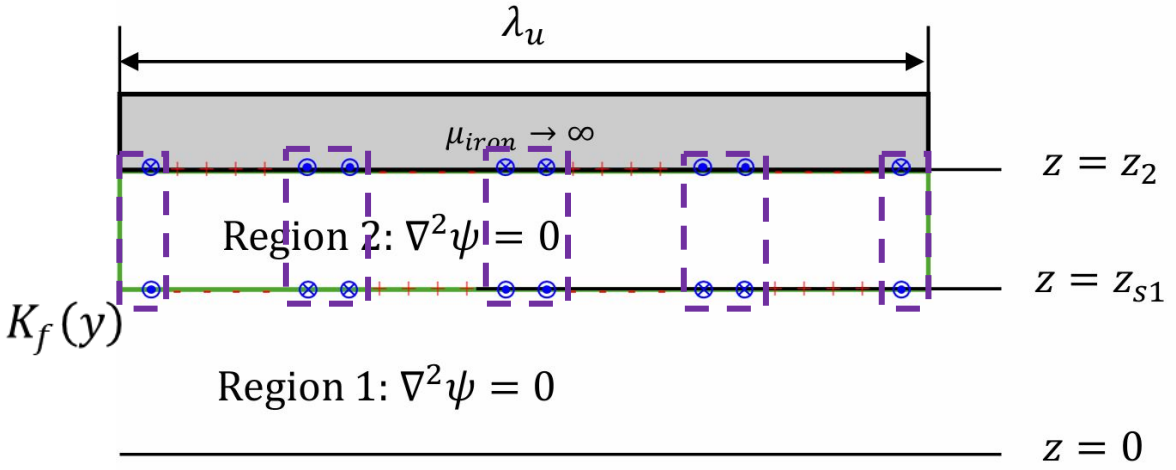




Fourier series approach to multi-magnet configuration

Fourier series methods also discussed by Jansen (2007), Cho and Jun (2002), Cho et al (2001), Trumper et al (1996)





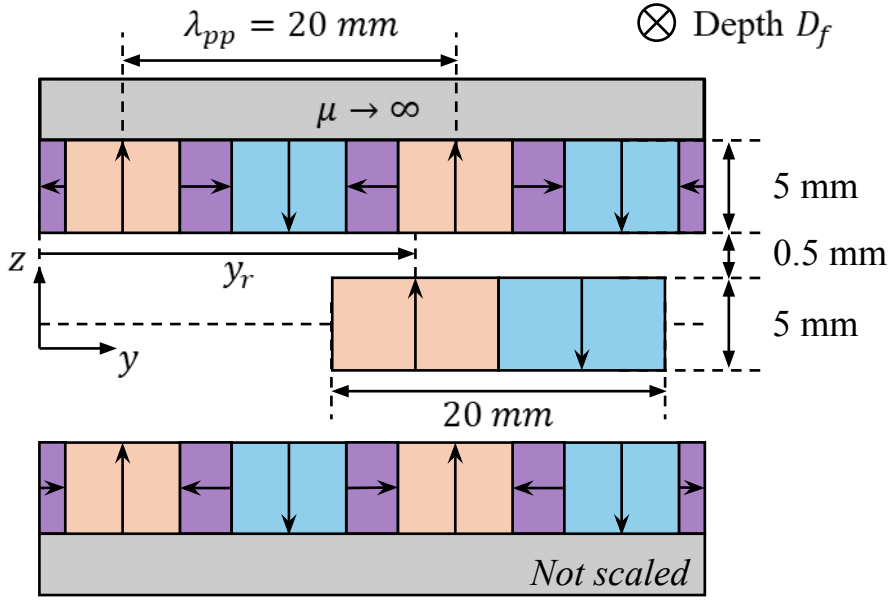
$$\sigma_m(y, y_r) = \sum_{k=1}^{\infty} a_k(y_r)\cos pky + b_k(y_r)\sin pky \ : \textit{Magnetic charge density of moving PM}$$

$H_{y1}(y, z_{m1})$: Fixed PM-driven magnetic intensity, evaluated at $z = z_{m1}$

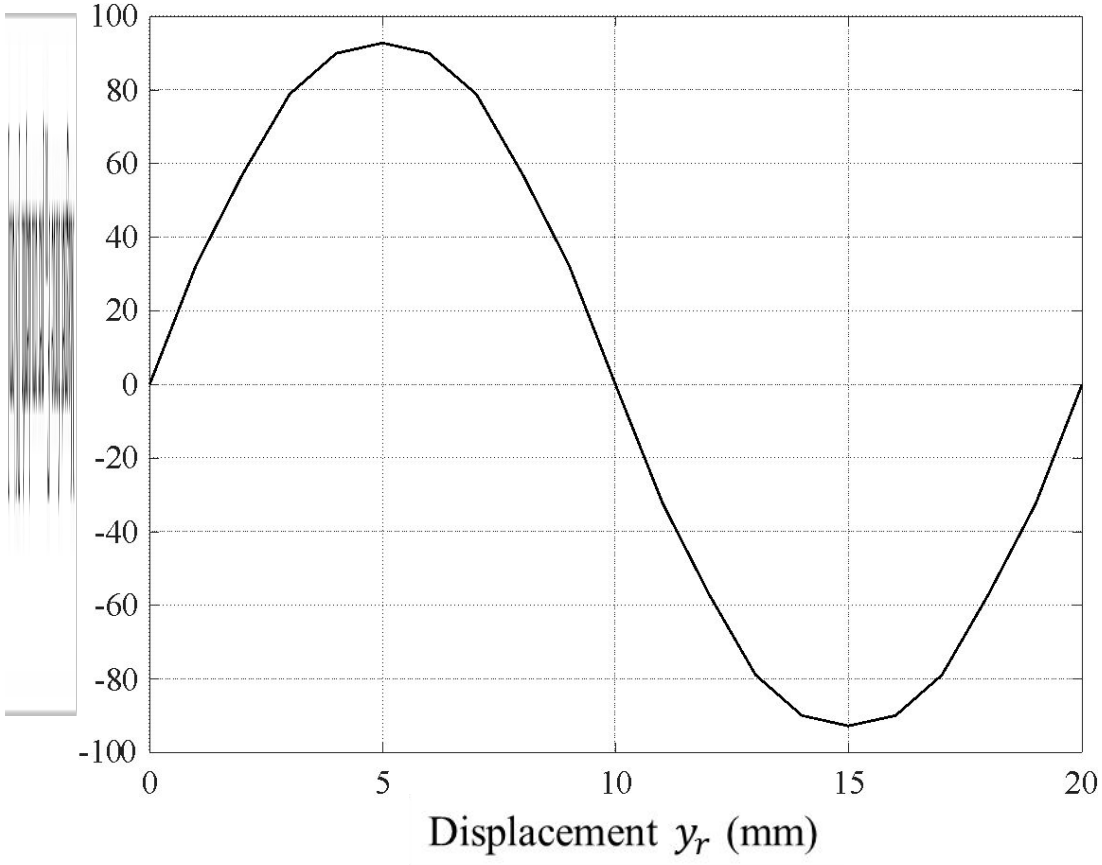
$$p = \frac{2\pi}{\lambda_u} : \textit{Fundamental spatial harmonic}$$

k: *Fourier harmonic number*

Fourier series approach to multi-magnet configuration



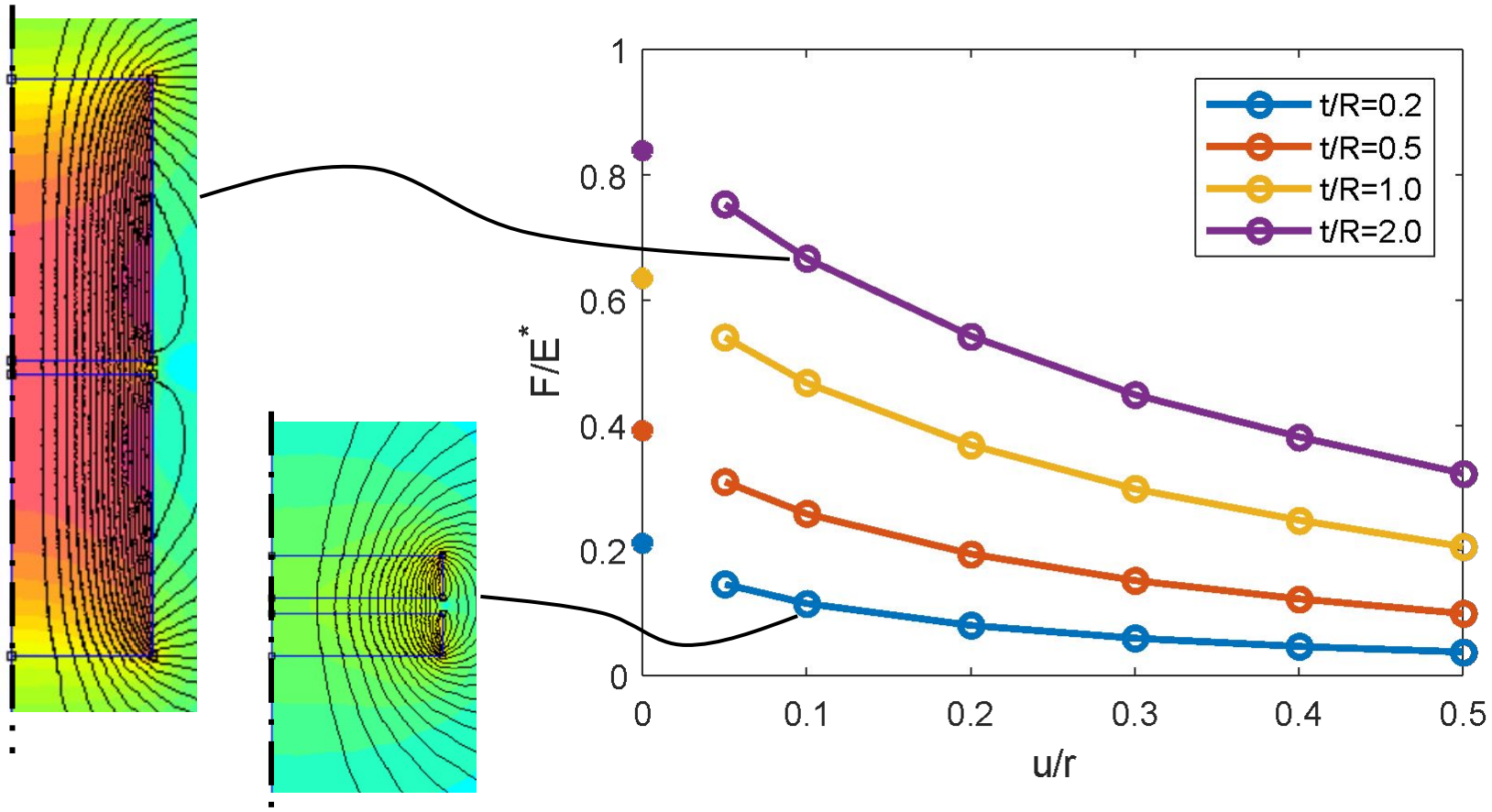
$$F_m(y_r) = D_f \int_0^{\lambda_u} \sigma_m(y, y_r)\, H_{y1}(y, z_{m1})\, dy$$
$$= \frac{\mu_0 D_f \lambda_u}{2} \sum_{k=1}^{\infty} a_k(y_r) \frac{(1 - \cosh pk(z_2 - z_{s1}))A_k + \sinh pk(z_2 - z_{s1})B_k}{\sinh pkz_2}$$

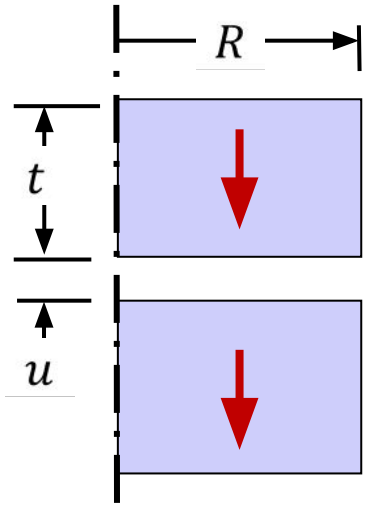


First harmonic is good to within a few percent

Normal Force Between Axially Poled Magnets

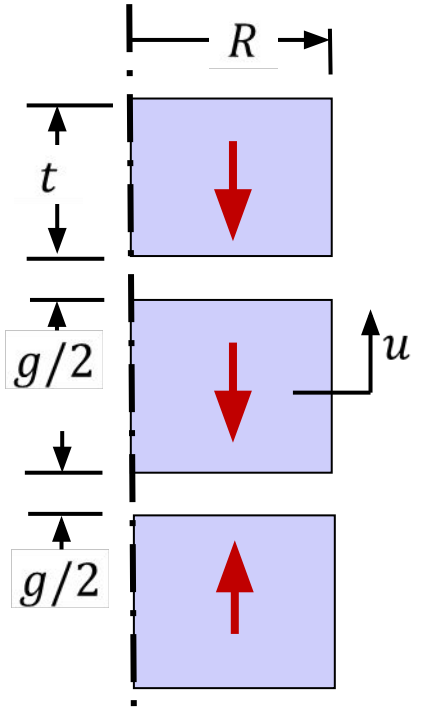
- For tall magnets ($t/R \gg 1$), the sticking force approaches E^*.
- If one of the magnets is flipped, they repel with almost the same magnitude force.

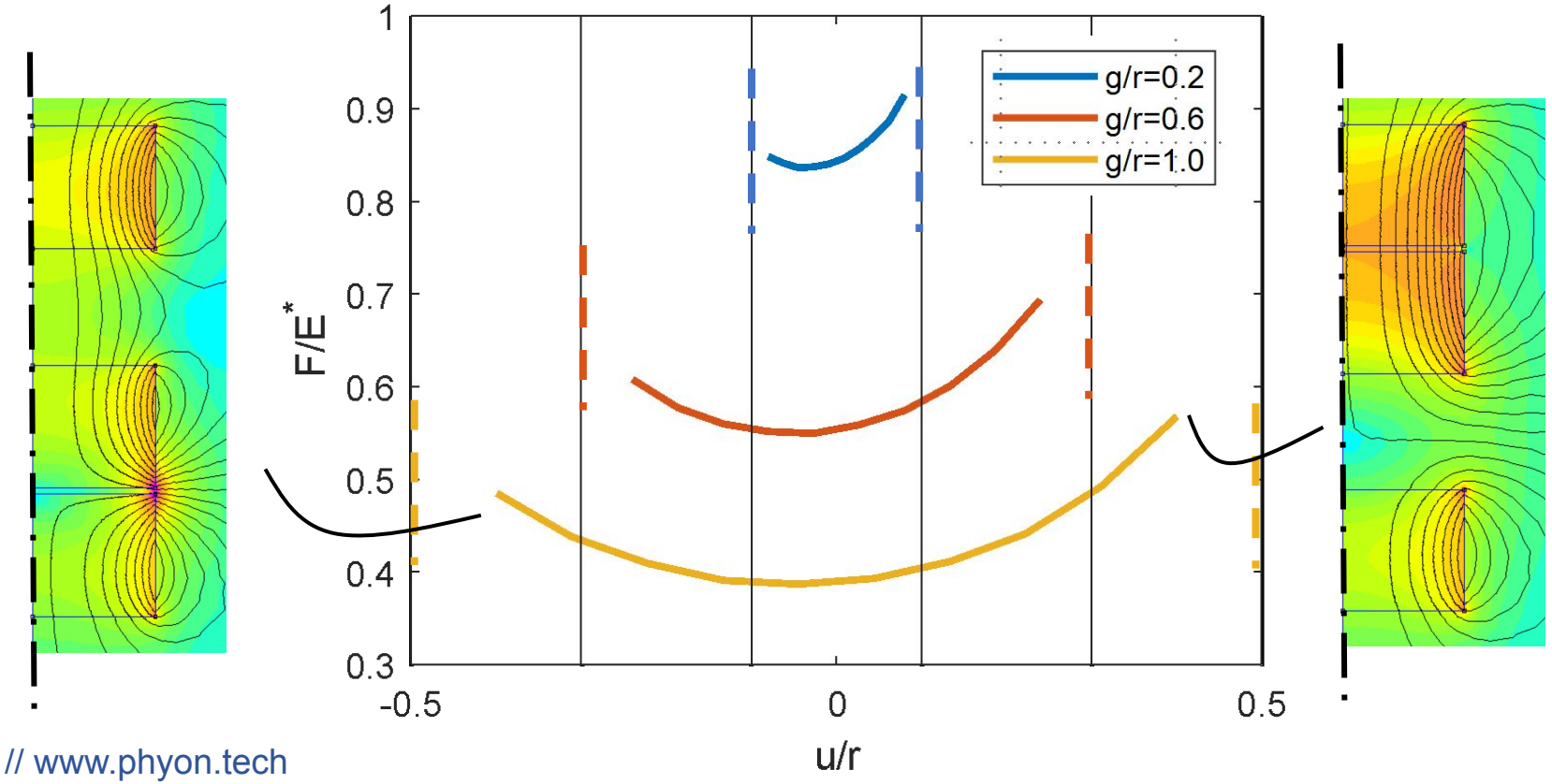




Push-Pull Normal Direction Springs

- Suppose we fix $t/R = 0.5$, and arrange three in push/pull arrangement.
- The force-displacement curve is flat near the center of travel.
- For small gaps and travels, we can attain $F/E^* \approx 0.9$.





Conclusions and design guidelines

- E* gives an approximate upper bound
- Change in energy is a good way to calculate force.
- Using 2D FEA is a useful tool to take advantage of planar configurations and symmetries.

- Next steps – design guidelines for:
  - Stiffness and parasitic forces
  - Misalignments
  - Temperature variation
  - Halbach magnets
  - Iron vs ironless
  - Hysteresis estimates
  - Adding an actuator